

2

Integration

2.1 Measurable transformations

Oftentimes, one is not interested in the full details of a measure space $(\Omega, \mathcal{F}, \mu)$ but only in certain functions defined on Ω . For example, if Ω represents the outcomes of 10 tosses of a fair coin, one may only be interested in knowing the number of heads in the 10 tosses. It turns out that to assign measures (probabilities) to sets (events) involving such functions, one can allow only certain functions (called measurable functions) that satisfy some ‘natural’ restrictions, specified in the following definitions.

Definition 2.1.1: Let Ω be a nonempty set and let $\mathcal{F}$ be a σ -algebra on Ω . Then the pair $(\Omega, \mathcal{F})$ is called a *measurable space*. If μ is a measure on $(\Omega, \mathcal{F})$, then the triplet $(\Omega, \mathcal{F}, \mu)$ is called a *measure space*. If in addition, μ is a probability measure, then $(\Omega, \mathcal{F}, \mu)$ is called a *probability space*.

Definition 2.1.2:

- (a) Let $(\Omega, \mathcal{F})$ be a measurable space. Then a function $f : \Omega$ to $\mathbb{R}$ is called $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -*measurable* (or $\mathcal{F}$ -*measurable*) if for each a in $\mathbb{R}$

$$f^{-1}((-\infty, a]) \equiv \{\omega : f(\omega) \leq a\} \in \mathcal{F}. \quad (1.1)$$

- (b) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Then a function $X : \Omega \rightarrow \mathbb{R}$ is called a *random variable*, if the event

$$X^{-1}((-\infty, a]) \equiv \{\omega : X(\omega) \leq a\} \in \mathcal{F}$$

for each a in $\mathbb{R}$, i.e., a random variable is a real valued $\mathcal{F}$ -measurable function on a probability space $(\Omega, \mathcal{F}, P)$.

It will be shown later that condition (1.1) on f is equivalent to the stronger condition that $f^{-1}(A) \in \mathcal{F}$ for all Borel sets $A \in \mathcal{B}(\mathbb{R})$. Since for any Borel set $A \in \mathcal{B}(\mathbb{R})$, $f^{-1}(A)$ is a member of the underlying σ -algebra $\mathcal{F}$, one can assign a measure to the set $f^{-1}(A)$ using a measure μ on $(\Omega, \mathcal{F})$. Note that for an *arbitrary* function T from $\Omega \rightarrow \mathbb{R}$, $T^{-1}(A)$ need not be a member of $\mathcal{F}$ and hence such an assignment may *not* be possible. Thus, condition (1.1) on real valued mappings is a ‘natural’ requirement while dealing with measure spaces.

The following definition generalizes (1.1) to maps between two measurable spaces.

Definition 2.1.3: Let $(\Omega_i, \mathcal{F}_i), i = 1, 2$ be measurable spaces. Then, a mapping $T : \Omega_1 \rightarrow \Omega_2$ is called *measurable with respect to the σ -algebras $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$* (or *$\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable*) if

$$T^{-1}(A) \in \mathcal{F}_1 \quad \text{for all } A \in \mathcal{F}_2.$$

Thus, X is a random variable on a probability space $(\Omega, \mathcal{F}, P)$ iff X is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Some examples of measurable transformations are given below.

Example 2.1.1: Let $\Omega = \{a, b, c, d\}$, $\mathcal{F}_2 = \{\Omega, \emptyset, \{a\}, \{b, c, d\}\}$ and let $\mathcal{F}_3$ be the set of all subsets of Ω . Define the mappings $T_i : \Omega \rightarrow \Omega, i = 1, 2$, by

$$T_1(\omega) \equiv a \text{ for } \omega \in \Omega$$

and

$$T_2(\omega) = \begin{cases} a & \text{if } \omega = a, b \\ c & \text{if } \omega = c, d. \end{cases}$$

Then, T_1 is $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$ -measurable since for any $A \in \mathcal{F}_3$, $T_1^{-1}(A) = \Omega$ or $\emptyset$ according as $a \in A$ or $a \notin A$. By similar arguments, it follows that T_2 is $\langle \mathcal{F}_3, \mathcal{F}_2 \rangle$ -measurable. However, T_2 is not $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$ -measurable since $T_2^{-1}(\{a\}) = \{a, b\} \notin \mathcal{F}_2$.

As this simple example shows, measurability of a given mapping critically depends on the σ -algebras on its domain and range spaces. In general, if T is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable, then T is $\langle \tilde{\mathcal{F}}_1, \mathcal{F}_2 \rangle$ -measurable for any σ -algebra $\tilde{\mathcal{F}}_1 \supset \mathcal{F}_1$ and T is $\langle \mathcal{F}_1, \tilde{\mathcal{F}}_2 \rangle$ -measurable for any $\tilde{\mathcal{F}}_2 \subset \mathcal{F}_2$.

Example 2.1.2: Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$T(x) = \begin{cases} \sin 2x & \text{if } x > 0 \\ 1 + \cos x & \text{if } x \leq 0. \end{cases}$$

Is T measurable w.r.t. the Borel σ -algebras $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$? If one is to apply the definition directly, one must check that $T^{-1}(A) \in \mathcal{B}(\mathbb{R})$ for all $A \in \mathcal{B}(\mathbb{R})$. However, finding $T^{-1}(A)$ for all Borel sets A is not an easy task. In many instances like this one, verification of the measurability property of a given mapping by directly using the definition can be difficult. In such situations, one may use some easy-to-verify sufficient conditions. Some results of this type are given below.

Proposition 2.1.1: *Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2, 3$ be measurable spaces.*

- (i) *Suppose that $\mathcal{F}_2 = \sigma\langle \mathcal{A} \rangle$ for some class of subsets $\mathcal{A}$ of Ω_2 . If $T : \Omega_1 \rightarrow \Omega_2$ is such that $T^{-1}(A) \in \mathcal{F}_1$ for all $A \in \mathcal{A}$, then T is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable.*
- (ii) *Suppose that $T_1 : \Omega_1 \rightarrow \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable and $T_2 : \Omega_2 \rightarrow \Omega_3$ is $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$ -measurable. Let $T = T_2 \circ T_1 : \Omega_1 \rightarrow \Omega_3$ denote the composition of T_1 and T_2 , defined by $T(\omega_1) = T_2(T_1(\omega_1))$, $\omega_1 \in \Omega_1$. Then, T is $\langle \mathcal{F}_1, \mathcal{F}_3 \rangle$ -measurable.*

Proof:

- (i) Define the collection of sets

$$\mathcal{F} = \{A \in \mathcal{F}_2 : T^{-1}(A) \in \mathcal{F}_1\}.$$

Then,

- (a) $T^{-1}(\Omega_2) = \Omega_1 \in \mathcal{F}_1 \Rightarrow \Omega_2 \in \mathcal{F}$.
- (b) If $A \in \mathcal{F}$, then $T^{-1}(A) \in \mathcal{F}_1 \Rightarrow (T^{-1}(A))^c \in \mathcal{F}_1 \Rightarrow T^{-1}(A^c) = (T^{-1}(A))^c \in \mathcal{F}_1$, implying $A^c \in \mathcal{F}$.
- (c) If $A_1, A_2, \dots \in \mathcal{F}$, then, $T^{-1}(A_i) \in \mathcal{F}_1$ for all $i \geq 1$. Since $\mathcal{F}_1$ is a σ -algebra, $T^{-1}(\bigcup_{n \geq 1} A_n) = \bigcup_{n \geq 1} T^{-1}(A_n) \in \mathcal{F}_1$. Thus, $\bigcup_{n \geq 1} A_n \in \mathcal{F}$. (See also Problem 2.1 on de Morgan's laws.)

Hence, by (a), (b), (c), $\mathcal{F}$ is a σ -algebra and by hypothesis $\mathcal{A} \subset \mathcal{F}$. Hence, $\mathcal{F}_2 = \sigma\langle \mathcal{A} \rangle \subset \mathcal{F} \subset \mathcal{F}_2$. Thus, $\mathcal{F} = \mathcal{F}_2$ and T is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable.

- (ii) Let $A \in \mathcal{F}_3$. Then, $T_2^{-1}(A) \in \mathcal{F}_2$, since T_2 is $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$ -measurable. Also, by the $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurability of T_1 , $T^{-1}(A) = T_1^{-1}(T_2^{-1}(A)) \in \mathcal{F}_1$, showing that T is $\langle \mathcal{F}_1, \mathcal{F}_3 \rangle$ -measurable. $\square$

Proposition 2.1.2: *For any $k, p \in \mathbb{N}$, if $f : \mathbb{R}^p \rightarrow \mathbb{R}^k$ is continuous, then f is $\langle \mathcal{B}(\mathbb{R}^p), \mathcal{B}(\mathbb{R}^k) \rangle$ -measurable.*

Proof: Let $\mathcal{A} = \{A : A \text{ is an open set in } \mathbb{R}^k\}$. Then, by the continuity of f , $f^{-1}(A)$ is open and hence, is in $\mathcal{B}(\mathbb{R}^p)$ (cf. Section A.4). Thus, $f^{-1}(A) \in$

$\mathbb{R}^p$ for all $A \in \mathcal{A}$. Since $\mathcal{B}(\mathbb{R}^k) = \sigma\langle \mathcal{A} \rangle$, by Proposition 2.1.1 (a), f is $\langle \mathcal{B}(\mathbb{R}^p), \mathcal{B}(\mathbb{R}^k) \rangle$ -measurable. $\square$

Although the converse to the above proposition is not true, a result due to Lusin says that except on a set of small measure, f coincides with a continuous function. This is stronger than the statement that except on set of small measure, f is close to a continuous function. For the statement and proof of Lusin's theorem, see Theorem 2.5.12.

Proposition 2.1.3: *Let $f_1, \dots, f_k$ ($k \in \mathbb{N}$) be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable transformations from Ω to $\mathbb{R}$. Then,*

- (i) $f = (f_1, \dots, f_k)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^k) \rangle$ -measurable.
- (ii) $g = f_1 + \dots + f_k$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
- (iii) $h \equiv \prod_{i=1}^k f_i$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
- (iv) Let $p \in \mathbb{N}$ and let $\psi : \mathbb{R}^k \rightarrow \mathbb{R}^p$ be continuous. Then, $\xi \equiv \psi \circ f$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^p) \rangle$ -measurable, where $f = (f_1, \dots, f_k)$.

Proof: To prove (i), note that for any rectangle $R = (a_1, b_1) \times \dots \times (a_k, b_k)$,

$$\begin{aligned} f^{-1}(R) &= \{\omega \in \Omega : a_1 < f_1(\omega) < b_1, \dots, a_k < f_k(\omega) < b_k\} \\ &= \bigcap_{i=1}^k \{\omega \in \Omega : a_i < f_i(\omega) < b_i\} \\ &= \bigcap_{i=1}^k f_i^{-1}(a_i, b_i) \in \mathcal{F}, \end{aligned}$$

since each f_i is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Hence, by Proposition 2.1.1 (i), f is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^k) \rangle$ -measurable. To prove (ii), note that the function $g_1(x) \equiv x_1 + \dots + x_k$, $x = (x_1, \dots, x_k) \in \mathbb{R}^k$ is continuous on $\mathbb{R}^k$, and hence, by Proposition 2.1.2, is $\langle \mathcal{B}(\mathbb{R}^k), \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Since $g = g_1 \circ f$, g is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, by Proposition 2.1.1 (ii). The proofs of (iii) and (iv) are similar to that of (ii) and hence, are omitted. $\square$

Corollary 2.1.4: *The collection of $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable functions from Ω to $\mathbb{R}$ is closed under pointwise addition and multiplication as well as under scalar multiplication.*

The proof of Corollary 2.1.4 is omitted.

In view of the above, writing the function T of Example 2.1.2 as

$$T(x) = (\sin 2x)I_{(0, \infty)}(x) + (1 + \cos x)I_{(-\infty, 0]}(x),$$

$x \in \mathbb{R}$, the $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurability of T follows. Note that here T is not continuous over $\mathbb{R}$ but only piecewise continuous (see also Problem 2.2).

Next, measurability of the limit of a sequence of measurable functions is considered. Let $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$ denote the extended real line and let $\mathcal{B}(\bar{\mathbb{R}}) \equiv \sigma(\mathcal{B}(\mathbb{R}) \cup \{+\infty\} \cup \{-\infty\})$ denote the extended Borel σ -algebra on $\bar{\mathbb{R}}$.

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Proposition 2.1.5: For each $n \in \mathbb{N}$, let $f_n : \Omega \rightarrow \bar{\mathbb{R}}$ be a $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable function.

- (i) Then, each of the functions $\sup_{n \in \mathbb{N}} f_n$, $\inf_{n \in \mathbb{N}} f_n$, $\limsup_{n \rightarrow \infty} f_n$, and $\liminf_{n \rightarrow \infty} f_n$ is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable.
- (ii) The set $A \equiv \{\omega : \lim_{n \rightarrow \infty} f_n(\omega) \text{ exists and is finite}\}$ lies in $\mathcal{F}$ and the function $h \equiv (\lim_{n \rightarrow \infty} f_n) \cdot I_A$ is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable.

Proof:

- (i) Let $g = \sup_{n \geq 1} f_n$. To show that g is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable, it is enough to show that $\{\omega : g(\omega) \leq r\} \in \mathcal{F}$ for all $r \in \mathbb{R}$ (cf. Problem 2.4). Now, for any $r \in \mathbb{R}$,

$$\begin{aligned} \limsup (f_n) &= \inf_{n \geq 1} \left(\sup_{m \geq n} f_m \right) \\ \{\omega : g(\omega) \leq r\} &= \bigcap_{n=1}^{\infty} \{\omega : f_n(\omega) \leq r\} \quad \leftarrow \text{like max is mble } f_n \text{ proof} \\ &= \bigcap_{n=1}^{\infty} f_n^{-1}((-\infty, r]) \in \mathcal{F}, \end{aligned}$$

since $f_n^{-1}((-\infty, r]) \in \mathcal{F}$ for all $n \geq 1$, by the measurability of f_n .

Next note that $\inf_{n \geq 1} f_n = -\sup_{n \geq 1} (-f_n)$ and hence, $\inf_{n \geq 1} f_n$ is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable. To prove the measurability of $\limsup_{n \rightarrow \infty} f_n$, define the functions $g_m = \sup_{n \geq m} f_n$, $m \geq 1$. Then, g_m is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable for each $m \geq 1$ and since g_m is nonincreasing in m , $\inf_{m \geq 1} g_m \equiv \limsup_{n \rightarrow \infty} f_n$ is also $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable. A similar argument works for $\liminf_{n \rightarrow \infty} f_n$.

- (ii) Let $h_1 = \limsup_{n \rightarrow \infty} f_n$ and $h_2 = \liminf_{n \rightarrow \infty} f_n$, and define $\tilde{h}_i = h_i I_{\mathbb{R}}(h_i)$, $i = 1, 2$. Note that $\tilde{h}_1 - \tilde{h}_2$ is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable. Hence,

$$\begin{aligned} &\overbrace{h_1, \text{ so long as values are finite}} \quad \{\omega : \lim_{n \rightarrow \infty} f_n(\omega) \text{ exists and is finite}\} \\ &= \{\omega : -\infty < \limsup_{n \rightarrow \infty} f_n(\omega) = \liminf_{n \rightarrow \infty} f_n(\omega) < \infty\} \\ &= \{\omega : -\infty < h_2(\omega) = h_1(\omega) < \infty\} \quad \begin{matrix} \swarrow h_1 \text{ finite} \\ \searrow h_2 \text{ finite} \end{matrix} \\ &= \{\omega : \tilde{h}_1(\omega) = \tilde{h}_2(\omega)\} \cap \{\omega : h_1(\omega) < \infty, h_2(\omega) > -\infty\} \\ &= (\tilde{h}_1 - \tilde{h}_2)^{-1}(\{0\}) \cap \{\omega : h_1(\omega) < \infty\} \cap \{\omega : h_2(\omega) > -\infty\} \in \mathcal{F}. \end{aligned}$$

$$\{h_1 < \infty\} = \bigcup_{n \geq 1} \{h_1 < n\} \quad \cap \quad \{h_2 > -\infty\}$$

Finally, note that $h = h_1 I_A$.

□

Definition 2.1.4: Let $\{f_\lambda : \lambda \in \Lambda\}$ be a family of mappings from Ω_1 into Ω_2 and let $\mathcal{F}_2$ be a σ -algebra on Ω_2 . Then,

is called the σ -algebra generated by $\{f_\lambda : \lambda \in \Lambda\}$ (w.r.t. $\mathcal{F}_2$) and is denoted by $\sigma\langle\{f_\lambda : \lambda \in \Lambda\}\rangle$.

Handwritten notes: $\sigma\langle\{f_\lambda^{-1}(A) : A \in \mathcal{F}_2, \lambda \in \Lambda\}\rangle$ with a green arrow pointing to the definition. Above the definition, $f_\lambda^{-1}(A)$ is written with a green arrow pointing to the set $\{f_\lambda^{-1}(A) : A \in \mathcal{F}_2, \lambda \in \Lambda\}$. Below the definition, $\Omega_1, \mathcal{F}_1$ and $\Omega_2, \mathcal{F}_2$ are written with green arrows pointing to the spaces and algebras.

Note that $\sigma\langle\{f_\lambda : \lambda \in \Lambda\}\rangle$ is the smallest σ -algebra on Ω_1 that makes all f_λ 's measurable w.r.t. $\mathcal{F}_2$ on Ω_2 .

Example 2.1.3: Let $f = I_A$ for some set $A \subset \Omega_1$ and $\Omega_2 = \mathbb{R}$ and $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$. Then,

$$\sigma\langle\{f\}\rangle = \sigma\langle\{A\}\rangle = \{\Omega_1, \emptyset, A, A^c\}.$$

Example 2.1.4: Let $\Omega_1 = \mathbb{R}^k$, $\Omega_2 = \mathbb{R}$, $\mathcal{F}_2 = \mathcal{B}(\mathbb{R})$, and for $1 \leq i \leq k$, let $f_i : \Omega_1 \rightarrow \Omega_2$ be defined as

$$f_i(x_1, \dots, x_k) = x_i, \quad (x_1, \dots, x_k) \in \Omega_1 = \mathbb{R}^k.$$

Handwritten notes: "coordinate maps" in green. $k=2$ in green. $f_1(x,y)=x$ and $f_2(x,y)=y$ in green.

Then, $\sigma\langle\{f_i : 1 \leq i \leq k\}\rangle = \mathcal{B}(\mathbb{R}^k)$.

Handwritten notes: "Smallest σ -alg that makes all k coordinate maps mble is $\mathcal{B}(\mathbb{R}^k)$ " in green. "2-1" in blue with an arrow pointing to the right.

To show this, note that any measurable rectangle $A_1 \times \dots \times A_k$ can be written as $A_1 \times \dots \times A_k = \bigcap_{i=1}^k f_i^{-1}(A_i)$ and hence $A_1 \times \dots \times A_k \in \sigma\langle\{f_i : 1 \leq i \leq k\}\rangle$ for all $A_1, \dots, A_k \in \mathcal{B}(\mathbb{R})$. Since $\mathbb{R}^k$ is generated by the collection of all measurable rectangles, $\mathcal{B}(\mathbb{R}^k) \subset \sigma\langle\{f_i : 1 \leq i \leq k\}\rangle$. Conversely, for any $A \in \mathcal{B}(\mathbb{R})$ and for any $1 \leq i \leq k$, $f_i^{-1}(A) = \mathbb{R} \times \dots \times A \times \dots \times \mathbb{R}$ (with A in the i th position) is in $\mathcal{B}(\mathbb{R}^k)$. Therefore, $\sigma\langle\{f_i : 1 \leq i \leq k\}\rangle = \sigma\langle\{f_i^{-1}(A) : A \in \mathcal{B}(\mathbb{R}), 1 \leq i \leq k\}\rangle \subset \mathcal{B}(\mathbb{R}^k)$. Hence, $\sigma\langle\{f_i : 1 \leq i \leq k\}\rangle = \mathcal{B}(\mathbb{R}^k)$.

Handwritten notes: "A $\in \mathbb{R}$ " in blue. $f_1^{-1}(A) = \{(x,y) : f_1(x,y) \in A\}$ in blue. $A \times \mathbb{R}$ in blue. $B \in \mathcal{B}(\mathbb{R})$ in blue. $f_2^{-1}(B) = \mathbb{R} \times B$ in blue. $A \times B = f_1^{-1}(A) \cap f_2^{-1}(B)$ in blue.

Proposition 2.1.6: Let $\{f_\lambda : \lambda \in \Lambda\}$ be an uncountable collection of maps from Ω_1 to Ω_2 . Then for any $B \in \sigma\langle\{f_\lambda : \lambda \in \Lambda\}\rangle$, there exists a countable set $\Lambda_B \subset \Lambda$ such that $B \in \sigma\langle\{f_\lambda : \lambda \in \Lambda_B\}\rangle$.

Proof: The proof of this result is left as an exercise (Problem 2.5). $\square$

2.2 Induced measures, distribution functions

Suppose X is a random variable defined on a probability space $(\Omega, \mathcal{F}, P)$. Then P governs the probabilities assigned to events like $X^{-1}([a, b])$, $-\infty < a < b < \infty$. Since X takes values in the real line, one should be able to express such probabilities only as a function of the set $[a, b]$. Clearly, since X is $(\mathcal{F}, \mathbb{R})$ -measurable, $X^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{B}(\mathbb{R})$ and the function

$$P_X(A) \equiv P(X^{-1}(A)) \quad (2.1)$$

is a set function defined on $\mathcal{B}(\mathbb{R})$. Is this a (probability) measure on $\mathcal{B}(\mathbb{R})$? The following proposition answers the question more generally.

Proposition 2.2.1: Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be measurable spaces and let $T : \Omega_1 \rightarrow \Omega_2$ be a $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable mapping from Ω_1 to Ω_2 . Then, for any measure μ on $(\Omega_1, \mathcal{F}_1)$, the set function μT^{-1} , defined by

$$\mu T^{-1}(A) = \mu T^{-1}(A) \equiv \mu(T^{-1}(A)), \quad A \in \mathcal{F}_2 \quad (2.2)$$

is a measure on $\mathcal{F}_2$.

Proof: It is easy to check that μT^{-1} satisfies the three conditions for being a measure. The details are left as an exercise (cf. Problem 2.9).

Definition 2.2.1: The measure μT^{-1} is called the *measure induced by T* (or the *induced measure of T*) on $\mathcal{F}_2$.

In particular, if $\mu(\Omega_1) = 1$, then $\mu T^{-1}(\Omega_2) = 1$. Hence, the set function P defined in (2.1) is indeed a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 2.2.2: For a random variable X defined on a probability space $(\Omega, \mathcal{F}, P)$, the *probability distribution* of X (or the *law* of X), denoted by P_X (say), is the induced measure of X under P on $\mathbb{R}$, as defined in (2.2).

In introductory courses on probability and statistics, one defines probabilities of events like ' $X \in [a, b]$ ' by using the probability mass function for discrete random variables and the probability density function for 'continuous' random variables. The measure-theoretic definition above allows one to treat both these cases as well as the case of 'mixed' distributions under a unified framework.

Definition 2.2.3: The *cumulative distribution function* (or *cdf* in short) of a random variable X is defined as

$$P(X \leq x) = F_X(x) \equiv P_X((-\infty, x]), \quad x \in \mathbb{R}. \quad (2.3)$$

Proposition 2.2.2: Let F be the cdf of a random variable X .

- (i) For $x_1 < x_2$, $F(x_1) \leq F(x_2)$ (i.e., F is nondecreasing on $\mathbb{R}$).
- (ii) For x in $\mathbb{R}$, $F(x) = \lim_{y \downarrow x} F(y)$ (i.e., F is right continuous on $\mathbb{R}$).
- (iii) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$.

$X \rightarrow F_X$

Proof: For $x_1 < x_2$, $(-\infty, x_1] \subset (-\infty, x_2]$. Since P_X is a measure on $\mathcal{B}(\mathbb{R})$,

$$F(x_1) = P_X((-\infty, x_1]) \leq P_X((-\infty, x_2]) = F(x_2), \quad P_X \text{ is monotone (measure)}$$

proving (i).

To prove (ii), let $x_n \downarrow x$. Then, the sets $(-\infty, x_n] \downarrow (-\infty, x]$, and $P_X((-\infty, x_1]) = P(X \leq x_1) \leq 1$. Hence, using the monotone continuity from above of the measure P_X (m.c.f.a.) (cf. Proposition 1.2.3), one gets

$$\lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} P_X((-\infty, x_n]) = P_X((-\infty, x]) = F(x).$$

Next consider part (iii). Note that if $x_n \downarrow -\infty$ and $y_n \uparrow \infty$, then $(-\infty, x_n] \downarrow \emptyset$ and $(-\infty, y_n] \uparrow (-\infty, \infty)$. Hence, part (iii) follows the m.c.f.a. and the m.c.f.b. properties of P_X (cf. Propositions 1.2.1 and 1.2.3). $\square$

Definition 2.2.4: A function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying (i), (ii), and (iii) of Proposition 2.2.2 is called a *cumulative distribution function* (or *cdf* for short).

Thus, given a random variable X , its cdf F_X satisfies properties (i), (ii), (iii) of Proposition 2.2.2. Conversely, given a cdf F , one can construct a probability space $(\Omega, \mathcal{F}, P)$ and a random variable X on it such that the cdf of X is F . Indeed, given a cdf F , note that by Theorem 1.3.3 and Definition 1.3.7, there exists a (Lebesgue-Stieltjes) probability measure μ_F on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mu_F((-\infty, x]) = F(x)$. Now define X to be the identity map on $\mathbb{R}$, i.e., let $X(x) \equiv x$ for all $x \in \mathbb{R}$. Then, X is a random variable on the probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$ with probability distribution $P_X = \mu_F$ and cdf $F_X = F$.

In addition to (i), (ii) and (iii) of Proposition 2.2.2, it is easy to verify that for any x in $\mathbb{R}$,

$$P(X < x) = F_X(x-) \equiv \lim_{y \uparrow x} F_X(y)$$

and hence

$$P(X = x) = F_X(x) - F_X(x-). \quad (2.4)$$

Thus, the function $F_X(\cdot)$ has a jump at x iff $P(X = x) > 0$. Since a monotone function from $\mathbb{R}$ to $\mathbb{R}$ can have only jump discontinuities and only a countable number of them (cf. Problem 2.11), for any random variable X , the set $\{a \in \mathbb{R} : P(X = a) > 0\}$ is countable. This leads to the following definitions.

Definition 2.2.5:

(or F_X has jump points in A)

- (a) A random variable X is called *discrete* if there exists a countable set $A \subset \mathbb{R}$ such that $P(X \in A) = 1$.
- (b) A random variable X is called *continuous* if $P(X = x) = 0$ for all $x \in \mathbb{R}$.

(F_X is a continuous fn)

Note that X is continuous iff F_X is continuous on all of $\mathbb{R}$ and X is discrete iff the sum of all the jumps of its cdf F_X is one. It may also be

$$(-\infty, x_n] \supset (-\infty, x_{n+1}]$$

$$\text{since } x_n \geq x_{n+1}$$

$$\& (-\infty, x_n] \in \mathcal{B}(\mathbb{R})$$

$\Rightarrow P$ has m.c.f.a

$$F \rightarrow X \text{ s.t. } F_X = F$$

given F , show
 $\exists X$ on $(\Omega, \mathcal{F}, P)$
 s.t. $F_X = F$

$$\begin{aligned} & \{a : P(X=a) > 0\} \\ &= \bigcup_{n \in \mathbb{N}} \{a : P(X=a) > \frac{1}{n}\} \\ &= \end{aligned}$$

need not be
 continuous or
 discrete

noted that if F_X is a step function, then X is discrete but not conversely. For example, consider the case where the set A in the above definition is the set of all rational numbers.

It turns out that a given cdf may be written as a weighted sum of a discrete and a continuous cdfs. Let F be a cdf. Let $A \equiv \{x : p(x) \equiv F(x) - F(x-) > 0\}$. As remarked earlier, A is at most countable. Write $\alpha = \sum_{y \in A} p(y)$ and let $\tilde{F}_d(x) = \sum_{y \in A} p(y) I_{(-\infty, x]}(y)$, and $\tilde{F}_c(x) = F(x) - \tilde{F}_d(x)$. It is easy to verify that $\tilde{F}_c(\cdot)$ is continuous on $\mathbb{R}$. If $\alpha = 0$, then $F(x) = \tilde{F}_c(x)$ and F is continuous. If $\alpha = 1$, then $F = \tilde{F}_d(x)$ and F is discrete. If $0 < \alpha < 1$, $F(\cdot)$ can be written as

$$F(x) = \alpha \tilde{F}_d(x) + (1 - \alpha) \tilde{F}_c(x), \quad (2.5)$$

where $F_d(x) \equiv \alpha^{-1} \tilde{F}_d(x)$ and $F_c(x) \equiv (1 - \alpha)^{-1} \tilde{F}_c(x)$ are both cdfs, with F_d being discrete and F_c being continuous. For a further decomposition of $F_c(\cdot)$ into absolutely continuous and singular continuous components, see Chapter 4.

2.2.1 Generalizations to higher dimensions

Induced distributions of random vectors and the associated cdfs are briefly considered in this section. Let $X = (X_1, X_2, \dots, X_k)$ be a k -dimensional random vector defined on a probability space $(\Omega, \mathcal{F}, P)$. The probability distribution P_X of X is the induced probability measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$, defined by (cf. (2.1))

$$P_X(A) \equiv P(X^{-1}(A)) \quad A \in \mathcal{B}(\mathbb{R}^k). \quad (2.6)$$

The cdf F_X of X is now defined by

$$F_X(x) = P(X \leq x), \quad x \in \mathbb{R}^k, \quad (2.7)$$

where for any $x = (x_1, x_2, \dots, x_k)$ and $y = (y_1, y_2, \dots, y_k)$ in $\mathbb{R}^k$, $x \leq y$ means that $x_i \leq y_i$ for all $i = 1, \dots, k$.

The extension of Proposition 2.2.2 to the k -dimensional case is notationally involved. Here, an analog of Proposition 2.2.2 for the bivariate case, i.e., for $k = 2$ is stated.

Proposition 2.2.3: *Let F be the cdf of a bivariate random vector $X = (X_1, X_2)$.*

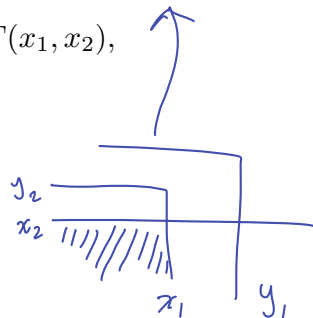
(i) *Then, for any $x = (x_1, x_2) \leq y = (y_1, y_2)$,*

$$F(y_1, y_2) - F(x_1, y_2) - F(y_1, x_2) + F(x_1, x_2) \geq 0. \quad (2.8)$$

(ii) *For any $x = (x_1, x_2) \in \mathbb{R}^2$,*

$$\lim_{y_1 \downarrow x_1, y_2 \downarrow x_2} F(y_1, y_2) = F(x_1, x_2),$$

i.e., F is right continuous on $\mathbb{R}^2$.



$\alpha = \sum_{y \in A} p(y) = 1/3$
in ④ of written notes

Cont. part in ④ of written notes

- (iii) $\lim_{x_1 \rightarrow -\infty} F(x_1, a) = \lim_{x_2 \rightarrow -\infty} F(a, x_2) = 0$ for all $a \in \mathbb{R}$;
 $\lim_{x_1 \rightarrow \infty, x_2 \rightarrow \infty} F(x_1, x_2) = 1$.
- (iv) For any $a \in \mathbb{R}$, $\lim_{y \uparrow \infty} F(a, y) = P(X_1 \leq a)$ and $\lim_{y \uparrow \infty} F(y, a) = P(X_2 \leq a)$.

Proof: Clearly,

$$\begin{aligned}
 0 &\leq P(X \in (x_1, y_1] \times (x_2, y_2]) \\
 &= P(x_1 < X_1 \leq y_1, x_2 < X_2 \leq y_2) \\
 &= P(X_1 \leq y_1, x_2 < X_2 \leq y_2) - P(X_1 \leq x_1, x_2 < X_2 \leq y_2) \\
 &= P(X_1 \leq y_1, X_2 \leq y_2) - P(X_1 \leq y_1, X_2 \leq x_2) \\
 &\quad - [P(X_1 \leq x_1, X_2 \leq y_2) - P(X_1 \leq x_1, X_2 \leq x_2)] \\
 &= F(y_1, y_2) - F(y_1, x_2) - F(x_1, y_2) + F(x_1, x_2).
 \end{aligned}$$

This proves (i).

To prove (ii), note that for any sequence $y_{in} \downarrow x_i, i = 1, 2$, the sets $A_n = (-\infty, y_{1n}] \times (-\infty, y_{2n}] \downarrow A \equiv (-\infty, x_1] \times (-\infty, x_2]$. Hence, by m.c.f.a property of a probability measure,

$$F(y_{1n}, y_{2n}) = P(X \in A_n) \downarrow P(A) = F(x_1, x_2).$$

For (iii), note that $(-\infty, x_{1n}] \times (-\infty, a] \downarrow \emptyset$ for any sequence $x_{1n} \downarrow -\infty$ and for any $a \in \mathbb{R}$. Hence, again by the m.c.f.a. property,

$$F(x_{1n}, a) \downarrow 0 \quad \text{as } n \rightarrow \infty.$$

By similar arguments, $F(a, x_{2n}) \downarrow 0$ whenever $x_{2n} \downarrow -\infty$. To prove the last relation in (iii), apply the m.c.f.b. property to the sets $(-\infty, x_{1n}] \times (-\infty, x_{2n}] \uparrow \mathbb{R}^2$ for $x_{1n} \uparrow \infty, x_{2n} \uparrow \infty$.

The proof of part (iv) is similar. $\square$

Note that any function satisfying properties (i), (ii), (iii) of Proposition 2.2.3 determines a probability measure uniquely. This follows from the discussions in Section 1.3, as (1.3.9) and (1.3.10) follow from (i) and (iii) (Problem 2.12). For a general $k \geq 1$, an analog of property (i) above is cumbersome to write down explicitly. Indeed, for any $x \leq y$, now a sum involving 2^k -terms must be nonnegative. However, properties (ii), (iii), and (iv) can be extended in an obvious way to the k -dimensional case. See Problem 2.13 for a precise statement. Also, for a general $k \geq 1$, functions satisfying the properties listed in Problem 2.13 uniquely determine a probability measure on $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$.

2.3 Integration (expectation of R.V.)

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f : \Omega \rightarrow \mathbb{R}$ be a measurable function. The goal of this section is to define the integral of f with respect to the

measure μ and establish some basic convergence results. The integral of a nonnegative function taking only *finitely* many values is defined first, which is then extended to all nonnegative measurable functions by approximation from below. Finally, the integral of an arbitrary measurable function is defined using the decomposition of the function into its positive and negative parts.

Definition 2.3.1: A function $f : \Omega \rightarrow \bar{\mathbb{R}} \equiv [-\infty, \infty]$ is called **simple** if there exist a finite set (of distinct elements) $\{c_1, \dots, c_k\} \in \bar{\mathbb{R}}$ and sets $A_1, \dots, A_k \in \mathcal{F}$, $k \in \mathbb{N}$ such that f can be written as

$$f = \sum_{i=1}^k c_i I_{A_i}. \quad (3.1)$$

will only consider
 $0 < c_i \leq \infty$

measurable only if $A_i \in \mathcal{F}$
 $= f^{-1}([c_i d_n, (j+1)d_n)) \in \mathcal{F}$ since f -measurable

Definition 2.3.2: (The integral of a simple nonnegative function). Let $f : \Omega \rightarrow \bar{\mathbb{R}}_+ \equiv [0, \infty]$ be a simple nonnegative function on $(\Omega, \mathcal{F}, \mu)$ with the representation (3.1). The *integral* of f w.r.t. μ , denoted by $\int f d\mu$, is defined as

$$\int f d\mu \equiv \sum_{i=1}^k c_i \mu(A_i). \quad (3.2)$$

Here and in the following, the relation

$$0 \cdot \infty = 0$$

is adopted as a convention.

It may be verified that the value of the integral in (3.2) does not depend on the representation of f . That is, if f can be expressed as $f = \sum_{j=1}^l d_j I_{B_j}$ for some $d_1, \dots, d_l \in \bar{\mathbb{R}}_+$ (not necessarily distinct) and for some sets $B_1, \dots, B_l \in \mathcal{F}$, then $\sum_{i=1}^k c_i \mu(A_i) = \sum_{j=1}^l d_j \mu(B_j)$, so that the value of the integral remains unchanged (Problem 2.17). Also note that for the f in Definition 2.3.2,

$$0 \leq \int f d\mu \leq \infty.$$

Property 10)
is the integral well-defined?

The following proposition is an **easy consequence** of the definition and the above remark.

Proposition 2.3.1: Let f and g be two simple nonnegative functions on $(\Omega, \mathcal{F}, \mu)$. Then

- (i) (Linearity) For $\alpha \geq 0, \beta \geq 0$, $\int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu$. $E(\alpha X + \beta Y) = \alpha EX + \beta EY$
- (ii) (Monotonicity) If $f \geq g$ a.e. (μ), i.e., $\mu(\{\omega : \omega \in \Omega, f(\omega) < g(\omega)\}) = 0$, then $\int f d\mu \geq \int g d\mu$.

Properties (i)-(iii)

he wrote this down

(iii) If $f = g$ a.e. (μ), i.e., if $\mu(\{\omega : \omega \in \Omega, f(\omega) \neq g(\omega)\}) = 0$, then $\int f d\mu = \int g d\mu$.

Definition 2.3.3: (The integral of a nonnegative measurable function). Let $f : \Omega \rightarrow \bar{\mathbb{R}}_+$ be a nonnegative measurable function on $(\Omega, \mathcal{F}, \mu)$. The integral of f with respect to μ , also denoted by $\int f d\mu$, is defined as

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu, \quad (3.3)$$

where $\{f_n\}_{n \geq 1}$ is any sequence of nonnegative simple functions such that $f_n(\omega) \uparrow f(\omega)$ for all ω .

Note that by Proposition 2.3.1 (ii), the sequence $\{\int f_n d\mu\}_{n \geq 1}$ is nondecreasing, and hence the right side of (3.3) is well defined. That the right side of (3.3) is the same for all such approximating sequences of functions needs to be established and is the content of the following proposition. The proof of this proposition exploits in a crucial way the m.c.f.b. and the finite additivity of the set function μ (or, equivalently, the countable additivity of μ).

Proposition 2.3.2: Let $\{f_n\}_{n \geq 1}$ and $\{g_n\}_{n \geq 1}$ be two sequences of simple nonnegative measurable functions on $(\Omega, \mathcal{F}, \mu)$ to $\bar{\mathbb{R}}_+$ such that as $n \rightarrow \infty$, $f_n(\omega) \uparrow f(\omega)$ and $g_n(\omega) \uparrow g(\omega)$ for all $\omega \in \Omega$. Then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \lim_{n \rightarrow \infty} \int g_n d\mu. \quad (3.4)$$

Proof: Fix $N \in \mathbb{N}$ and $0 < \rho < 1$. It will now be shown that

$$\lim_{n \rightarrow \infty} \int f_n d\mu \geq \rho \int g_N d\mu. \quad (3.5)$$

Suppose that g_N has the representation $g_N \equiv \sum_{i=1}^k d_i I_{B_i}$. Let $D_n = \{\omega \in \Omega : f_n(\omega) \geq \rho g_N(\omega)\}$, $n \geq 1$. Since $f_n(\omega) \uparrow f(\omega)$ for all ω , $D_n \uparrow D \equiv \{\omega : f(\omega) \geq \rho g_N(\omega)\}$ (Problem 2.18 (b)). Also since $g_N(\omega) \leq f(\omega)$ and $0 < \rho < 1$, $D = \Omega$. Now writing $f_n = f_n I_{D_n} + f_n I_{D_n^c}$, it follows from Proposition 2.3.1 that

$$\begin{aligned} \int f_n d\mu &\geq \int f_n I_{D_n} d\mu \geq \rho \int g_N I_{D_n} d\mu \\ &= \rho \sum_{i=1}^k d_i \mu(B_i \cap D_n). \end{aligned} \quad (3.6)$$

By the m.c.f.b. property, for each $i \in \mathbb{N}$, $\mu(B_i \cap D_n) \uparrow \mu(B_i \cap \Omega) = \mu(B_i)$ as $n \rightarrow \infty$. Since the sequence $\{\int f_n d\mu\}_{n \geq 1}$ is nondecreasing, taking limits

in (3.6), yields (3.5). Next, letting $\rho \uparrow 1$ yields $\lim_{n \rightarrow \infty} \int f_n d\mu \geq \int g_N d\mu$ for each $N \in \mathbb{N}$ and hence,

$$\lim_{n \rightarrow \infty} \int f_n d\mu \geq \lim_{n \rightarrow \infty} \int g_n d\mu.$$

By symmetry, (3.4) follows and hence, the proof is complete. $\square$

Remark 2.3.1: It is easy to verify that Proposition 2.3.2 remains valid if $\{f_n\}_{n \geq 1}$ and $\{g_n\}_{n \geq 1}$ increase to f a.e. (μ) .
 for all

see pt. of MCT

Given a nonnegative measurable function f , one can always construct a nondecreasing sequence $\{f_n\}_{n \geq 1}$ of nonnegative simple functions such that $f_n(\omega) \uparrow f(\omega)$ for all $\omega \in \Omega$ in the following manner. Let $\{\delta_n\}_{n \geq 1}$ be a sequence of positive real numbers and let $\{N_n\}_{n \geq 1}$ be a sequence of positive integers such that as $n \rightarrow \infty$, $\delta_n \downarrow 0$, $N_n \uparrow \infty$ and $N_n \delta_n \uparrow \infty$. Further, suppose that the sequence $P_n \equiv \{j\delta_n : j = 0, 1, 2, \dots, N_n\}$ is nested, i.e., $P_n \subset P_{n+1}$ for each $n \geq 1$. Now set

$$f_n(\omega) = \begin{cases} j\delta_n & \text{if } j\delta_n \leq f(\omega) < (j+1)\delta_n, \quad j = 0, 1, 2, \dots, (N_n - 1) \\ N_n\delta_n & \text{if } f(\omega) \geq N_n\delta_n. \end{cases} \quad (3.7)$$

A specific choice of δ_n and N_n is given by $\delta_n = 2^{-n}$, $N_n = n2^n$.
 almost everywhere

Thus, with the above choice of $\{f_n\}_{n \geq 1}$ in the definition of the Lebesgue integral $\int f d\mu$ in (3.3), the range of f is subdivided into intervals of decreasing lengths. This is in contrast to the definition of the Riemann integral of f over a bounded interval, which is defined via subdividing the domain of f into finer subintervals.

EXAMPLE of $f_n \uparrow f$

Picture in notes

$$f_n(\omega) = \sum_{i=1}^N c_i 1_{A_i}$$

$$A_i = \{\omega : j\delta_n \leq f(\omega) < (j+1)\delta_n\}$$

$$\delta_n = \frac{1}{2^n}, N_n = n \cdot 2^n$$

Remark 2.3.2: In some cases it may be more appropriate to choose the approximating sequence $\{f_n\}_{n \geq 1}$ in a manner different from (3.7). For example, let $\Omega = \{\omega_i : i \geq 1\}$ be a countable set, $\mathcal{F} = \mathcal{P}(\Omega)$, the power set of Ω , and let μ be a measure on $(\Omega, \mathcal{F})$. Then any function $f : \Omega \rightarrow \mathbb{R}_+ \equiv [0, \infty)$ is measurable and the integral $\int f d\mu$ coincides with the sum $\sum_{i=1}^{\infty} f(\omega_i) \mu(\{\omega_i\})$ as can be seen by choosing the approximating sequence $\{f_n\}_{n \geq 1}$ as

$$f_n(\omega_i) = \begin{cases} f(\omega_i) & \text{for } i = 1, 2, \dots, n \\ 0 & \text{for } i > n. \end{cases}$$

Remark 2.3.3: The integral of a nonnegative measurable function can be alternatively defined as

$$\int f d\mu = \sup \left\{ \int g d\mu : g \text{ nonnegative and simple, } g \leq f \right\}.$$

The equivalence of this to (3.3) is seen as follows. Clearly the right side above, say, M is greater than or equal to $\int f d\mu$ as in (3.3). Conversely,

there exist a sequence $\{g_n\}_{n \geq 1}$ of simple nonnegative functions with $g_n \leq f$ for all $n \geq 1$ such that $\lim_n \int g_n d\mu$ equals the supremum M defined above. Now set $h_n = \max\{g_j : 1 \leq j \leq n\}$, $n \geq 1$. Now it can be verified that for each $n \geq 1$, h_n is nonnegative, simple, and satisfies $h_n \uparrow f$ and also that $\int h_n d\mu$ converges to M (Problem 2.19 (b)).

Corollary 2.3.3: *Let f and g be two nonnegative measurable functions on $(\Omega, \mathcal{F}, \mu)$. Then, the conclusions of Proposition 2.3.1 remain valid for such f and g .*

Proof: This follows from Proposition 2.3.1 for nonnegative simple functions and Definition 2.3.3. $\square$

The definition of the integral $\int f d\mu$ of a nonnegative measurable function f in (3.3) makes it possible to interchange limits and integration in a fairly routine manner. In particular, the following key result is a direct consequence of the definition.

Theorem 2.3.4: (*The monotone convergence theorem or MCT*). *Let $\{f_n\}_{n \geq 1}$ and f be nonnegative measurable functions on $(\Omega, \mathcal{F}, \mu)$ such that $f_n \uparrow f$ a.e. (μ) . Then*

$\exists \{g_n\}$ noneg simple $\uparrow f_n$

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu. \quad (3.8)$$

$$\lim_{n \rightarrow \infty} E(X_n) = E(\lim_{n \rightarrow \infty} X_n)$$

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$$

Remark 2.3.4: The important difference between (3.4) and (3.8) is that in (3.8), the f_n 's need not be simple.

Proof: It is similar to the proof of Proposition 2.3.2. Let $\{g_n\}_{n \geq 1}$ be a sequence of nonnegative simple functions on $(\Omega, \mathcal{F}, \mu)$ such that $g_n(\omega) \uparrow f(\omega)$ for all ω . By hypothesis, there exists a set $A \in \mathcal{F}$ such that $\mu(A^c) = 0$ and for ω in A , $f_n(\omega) \uparrow f(\omega)$. Fix $k \in \mathbb{N}$ and $0 < \rho < 1$. Let $D_n = \{\omega : \omega \in A, f_n(\omega) \geq \rho g_k(\omega)\}$, $n \geq 1$. Then, $D_n \uparrow D \equiv \{\omega : \omega \in A, f(\omega) \geq \rho g_k(\omega)\}$. Since $g_k(\omega) \leq f(\omega)$ for all ω , it follows that $D = A$. Now, by Corollary 2.3.3,

$$\int f_n d\mu \geq \int f_n I_{D_n} d\mu \stackrel{\text{as in last proof}}{\geq} \rho \int g_k I_{D_n} d\mu \quad \text{for all } n \geq 1.$$

$$D_n = \tilde{D}_n \cap A$$

$$\uparrow \tilde{D} \cap A$$

$$= \{\omega \in A : f(\omega) \geq \rho g_k(\omega)\}$$

By m.c.f.b., $\int g_k I_{D_n} d\mu \uparrow \int g_k I_A d\mu = \int g_k d\mu$ as $n \rightarrow \infty$, yielding

$$\liminf_{n \rightarrow \infty} \int f_n d\mu \geq \rho \int g_k d\mu$$

for all $0 < \rho < 1$ and all $k \in \mathbb{N}$. Letting $\rho \uparrow 1$ first and then $k \uparrow \infty$, from (3.3) one gets

$$\liminf_{n \rightarrow \infty} \int f_n d\mu \geq \int f d\mu.$$

By monotonicity (Corollary 2.3.3),

$$\int f_n d\mu \leq \int f d\mu \quad \text{for all } n \geq 1$$

$\limsup_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu$

and so the proof is complete. $\square$

Corollary 2.3.5: Let $\{h_n\}_{n \geq 1}$ be a sequence of nonnegative measurable functions on a measure space $(\Omega, \mathcal{F}, \mu)$. Then

$$\int \left(\sum_{n=1}^{\infty} h_n \right) d\mu = \sum_{n=1}^{\infty} \int h_n d\mu. \quad \leftarrow \text{MCT}$$

Proof: Let $f_n = \sum_{i=1}^n h_i, n \geq 1$, and let $f = \sum_{i=1}^{\infty} h_i$. Then, $0 \leq f_n \uparrow f$. By the MCT,

$$\text{MCT: } \int f_n d\mu \uparrow \int f d\mu.$$

But by Corollary 2.3.3,

$$\int f_n d\mu = \sum_{i=1}^n \int h_i d\mu.$$

Hence, the result follows. $\square$

Corollary 2.3.6: Let f be a nonnegative measurable function on a measurable space $(\Omega, \mathcal{F}, \mu)$. For $A \in \mathcal{F}$, let

$$\nu(A) \equiv \int f I_A d\mu, \quad A \in \mathcal{F}$$

Then, ν is a measure on $(\Omega, \mathcal{F})$.

Proof: Let $\{A_n\}_{n \geq 1}$ be a sequence of disjoint sets in $\mathcal{F}$. Let $h_n = f I_{A_n}$ for $n \geq 1$. Then by Corollary 2.3.5,

$$\begin{aligned} \nu\left(\bigcup_{n \geq 1} A_n\right) &= \int f I_{\left[\bigcup_{n \geq 1} A_n\right]} d\mu = \int f \cdot \left[\sum_{n=1}^{\infty} I_{A_n} \right] d\mu \\ &= \int \left[\sum_{n=1}^{\infty} h_n \right] d\mu = \sum_{n=1}^{\infty} \int h_n d\mu = \sum_{n=1}^{\infty} \nu(A_n). \end{aligned}$$

$\leftarrow$ uses property of indicators

$\square$

Remark 2.3.5: Notice that $\mu(A) = 0 \Rightarrow \nu(A) = 0$. In this case ν is said to be dominated by or absolutely continuous with respect to μ . The Radon-Nikodym theorem (see Chapter 4) provides a converse to this. That is, if ν and μ are two measures on a measurable space $(\Omega, \mathcal{F})$ such that

$\nu \ll \mu$
notation

We can show, in this case, $\nu(A) = \int_A f d\mu, A \in \mathcal{F}$ for some $f \geq 0$

(proof in ch. 4)

In this case we write $f(x) = \frac{d\nu}{d\mu}(x), x \in \Omega \rightarrow$ & this fn f is our p.m.f (discrete) or density for cont. r.v.s

dominated

help us find prob

ν is dominated by μ and μ is σ -finite, then there exists a nonnegative measurable function f such that $\nu(A) = \int f I_A d\mu$ for all A in $\mathcal{F}$. This f is called a *Radon-Nikodym derivative* (or a *density*) of ν with respect to μ and is denoted by $\frac{d\nu}{d\mu}$.

Theorem 2.3.7: (Fatou's lemma). Let $\{f_n\}_{n \geq 1}$ be a sequence of nonnegative measurable functions on $(\Omega, \mathcal{F}, \mu)$. Then $\liminf E(X_n) \geq E(\liminf X_n)$

$$\liminf_{n \rightarrow \infty} \int f_n d\mu \geq \int \liminf_{n \rightarrow \infty} f_n d\mu. \quad (3.9)$$

Proof: Let $g_n(\omega) \equiv \inf\{f_j(\omega) : j \geq n\}$. Then $\{g_n\}_{n \geq 1}$ is a sequence of nonnegative, nondecreasing measurable functions on $(\Omega, \mathcal{F}, \mu)$ such that $g_n(\omega) \uparrow g(\omega) \equiv \liminf_{n \rightarrow \infty} f_n(\omega)$. By the MCT,

$$\int g_n d\mu \uparrow \int g d\mu = \int (\liminf f_n) d\mu$$

But by monotonicity

$$\int f_n d\mu \geq \int g_n d\mu \text{ for each } n \geq 1, \Rightarrow \liminf \int f_n d\mu \geq \liminf \int g_n d\mu$$

and hence, (3.9) follows. $\square$

Remark 2.3.6: In (3.9), the inequality can be strict. For example, take $f_n = I_{[n, \infty)}$, $n \geq 1$, on the measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ where m is the Lebesgue measure. For another example, consider $f_n = n I_{[0, \frac{1}{n}]}$, $n \geq 1$, on the finite measure space $([0, 1], \mathcal{B}([0, 1]), m)$.

Definition 2.3.4: (The integral of a measurable function). Let f be a real valued measurable function on a measure space $(\Omega, \mathcal{F}, \mu)$. Let $f^+ = f I_{\{f \geq 0\}}$ and $f^- = -f I_{\{f < 0\}}$. The integral of f with respect to μ , denoted by $\int f d\mu$, is defined as

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu,$$

provided that at least one of the integrals on the right side is finite.

Remark 2.3.7: Note that both f^+ and f^- are nonnegative measurable functions and $f = f^+ - f^-$ and $|f| = f^+ + f^-$. Further, the integrals $\int f^+ d\mu$ and $\int f^- d\mu$ in Definition 2.3.4 are defined via Definition 2.3.3.

Definition 2.3.5: (Integrable functions). A measurable function f on a measure space $(\Omega, \mathcal{F}, \mu)$ is said to be *integrable* with respect to μ if $\int |f| d\mu < \infty$.

Since $|f| = f^+ + f^-$, it follows that f is *integrable* iff both f^+ and f^- are integrable, i.e., $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$. In the following, whenever

Reverse Fatou's Lemma:
Fatou's for $\limsup$ b/c $\limsup = \liminf$

check

the integral of f or its integrability is discussed, the measurability of f will be assumed to hold.

Remark on notation: The $\int f d\mu$ is also written as

$$\int_{\Omega} f(\omega) \mu(d\omega) \quad \text{and} \quad \int_{\Omega} f(\omega) d\mu(\omega).$$

Definition 2.3.6: Let f be a measurable function on a measure space $(\Omega, \mathcal{F}, \mu)$ and $A \in \mathcal{F}$. Then integral of f over A with respect to μ , denoted by $\int_A f d\mu$, is defined as

$$\int_A f d\mu \equiv \int f I_A d\mu, \quad (3.10)$$

provided the right side is well defined.

Definition 2.3.7: (L^p -spaces). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $0 < p \leq \infty$. Then $L^p(\Omega, \mathcal{F}, \mu)$ is defined as

$$\begin{aligned} L^p(\Omega, \mathcal{F}, \mu) &\equiv \{f : |f|^p \text{ is integrable with respect to } \mu\} \\ &= \left\{ f : \int |f|^p d\mu < \infty \right\} \quad \text{for } 0 < p < \infty, \end{aligned}$$

and $f \in L^p(\mu) \Rightarrow f \in L^q(\mu)$

$$L^\infty(\Omega, \mathcal{F}, \mu) \equiv \left\{ f : \mu(\{|f| > K\}) = 0 \text{ for some } K \in (0, \infty) \right\}.$$

The following is an extension of Proposition 2.3.1 to integrable functions.

Proposition 2.3.8: Let $f, g \in L^1(\Omega, \mathcal{F}, \mu)$. Then

$$(i) \int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu \text{ for any } \alpha, \beta \in \mathbb{R}.$$

$$(ii) f \geq g \text{ a.e. } (\mu) \Rightarrow \int f d\mu \geq \int g d\mu.$$

$$(iii) f = g \text{ a.e. } (\mu) \Rightarrow \int f d\mu = \int g d\mu.$$

Proof: It is easy to verify (Problem 2.32) that if $h = h_1 - h_2$ where h_1 and h_2 are nonnegative functions in $L^1(\Omega, \mathcal{F}, \mu)$, then h is also in $L^1(\Omega, \mathcal{F}, \mu)$ and

$$\int h d\mu = \int h_1 d\mu - \int h_2 d\mu. \quad (3.11)$$

Note that $h \equiv \alpha f + \beta g$ can be written as

$$\begin{aligned} &(\alpha^+ - \alpha^-)(f^+ - f^-) + (\beta^+ - \beta^-)(g^+ - g^-) \\ &= (\alpha^+ f^+ + \alpha^- f^- + \beta^+ g^+ + \beta^- g^-) \\ &\quad - (\alpha^+ f^- + \alpha^- f^+ + \beta^+ g^- + \beta^- g^+) \\ &= h_1 - h_2, \quad \text{say.} \end{aligned}$$

can show $L^p \subset L^q$ if $p < q$

all fn/R.V. w/ bounded support: e.g. unif(0,1), beta, counter e.g. N(0,1) gamma

integrable fns, 1st moment (integral is finite)

read proof briefly

$$\alpha \in \mathbb{R}, \alpha^+ = \alpha I_{(\alpha > 0)}, \alpha^- = (-\alpha) I_{(\alpha < 0)}$$

$$(+2)^+ = 2$$

$$(-2)^+ = 0$$

$$\alpha^+ - \alpha^- = \alpha$$

$$|\alpha| = \alpha^+ + \alpha^-$$

Since $f, g \in L^1(\Omega, \mathcal{F}, \mu)$, it follows that h_1 and $h_2 \in L^1(\Omega, \mathcal{F}, \mu)$. Further, they are nonnegative and by (3.11), $h \in L^1(\Omega, \mathcal{F}, \mu)$ and

$$\int h d\mu = \int h_1 d\mu - \int h_2 d\mu.$$

Now apply Proposition 2.3.1 to each of the terms on the right side and regroup the terms to get

$$\int h d\mu = \alpha \int f d\mu + \beta \int g d\mu.$$

Proofs of (ii) and (iii) are left as an exercise. $\square$

Remark 2.3.8: By Proposition 2.3.8, if f and $g \in L^1(\Omega, \mathcal{F}, \mu)$, then so does $\alpha f + \beta g$ for all $\alpha, \beta \in \mathbb{R}$. Thus, $L^1(\Omega, \mathcal{F}, \mu)$ is a vector space over $\mathbb{R}$. Further, if one sets

$$\|f\|_1 \equiv \int |f| d\mu,$$

$$\|f\|_p \equiv \left(\int |f|^p d\mu \right)^{1/p}$$

$$\|f\|_\infty \equiv \inf \{k : \mu(|f| > k) = 0\}$$

(and identifies a function f with its equivalence class under the relation $f \sim g$ iff $f = g$ a.e. (μ)), then $\|\cdot\|_1$ defines a norm on $L^1(\Omega, \mathcal{F}, \mu)$ and makes it a *normed linear space* (cf. Chapter 3). A similar remark also holds for $L^p(\Omega, \mathcal{F}, \mu)$ for $1 < p \leq \infty$.

Smallest k that bounds f (essential supremum)

Next note that by Proposition 2.3.8, if $f = 0$ a.e. (μ) , then $\int f d\mu = 0$. However, the converse is not true. But if f is nonnegative a.e. (μ) , then the converse is true as shown below.

Proposition 2.3.9: Let f be a measurable function on $(\Omega, \mathcal{F}, \mu)$ and let f be nonnegative a.e. (μ) . Then

$$\int f d\mu = 0 \iff f = 0 \text{ a.e. } (\mu).$$

Proof: It is enough to prove the “only if” part. Let $D = \{\omega : f(\omega) > 0\}$ and $D_n = \{\omega : f(\omega) > \frac{1}{n}\}$, $n \geq 1$. Then $D = \bigcup_{n \geq 1} D_n$. Since $f \geq f I_{D_n}$ a.e. (μ) ,

$$0 = \int f d\mu \geq \int f I_{D_n} d\mu \geq \frac{1}{n} \mu(D_n) \Rightarrow \mu(D_n) = 0 \text{ for each } n \geq 1.$$

Also $D_n \uparrow D$ and so by m.c.f.b.,

$$\mu(D) = \lim_{n \rightarrow \infty} \mu(D_n) = 0.$$

$$\Rightarrow \boxed{f = 0 \text{ a.e. } (\mu)}$$

Hence, Proposition 2.3.9 follows. $\square$

A dual to the above proposition is the next one.

Proposition 2.3.10: Let $f \in L^1(\Omega, \mathcal{F}, \mu)$. Then, $|f| < \infty$ a.e. (μ) .

Proof: Let $C_n = \{\omega : |f(\omega)| > n\}$, $n \geq 1$ and let $C = \{\omega : |f(\omega)| = \infty\}$. Then $C_n \downarrow C$ and

$$\infty > \int |f| d\mu \geq \int |f| I_{C_n} d\mu \geq n\mu(C_n) \Rightarrow \mu(C_n) \leq \frac{\int |f| d\mu}{n} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Since $\int |f| d\mu < \infty$, $\lim_{n \rightarrow \infty} \mu(C_n) = 0$. Hence, by m.c.f.a., $\mu(C) = \lim_{n \rightarrow \infty} \mu(C_n) = 0$. $\square$

The next result is a useful convergence theorem for integrals.

Theorem 2.3.11: (The extended dominated convergence theorem or EDCT). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f_n, g_n : \Omega \rightarrow \mathbb{R}$ be $(\mathcal{F}, \mathbb{R})$ -measurable functions such that $|f_n| \leq g_n$ a.e. (μ) for all $n \geq 1$. Suppose that

- (i) $g_n \rightarrow g$ a.e. (μ) and $f_n \rightarrow f$ a.e. (μ) ;
- (ii) $g_n, g \in L^1(\Omega, \mathcal{F}, \mu)$ and $\int |g_n| d\mu \rightarrow \int |g| d\mu$ as $n \rightarrow \infty$. Then, $f \in L^1(\Omega, \mathcal{F}, \mu)$,

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu \quad \text{and} \quad \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad (3.12)$$

Two important special cases of Theorem 2.3.11 will be stated next. When $g_n = g$ for all $n \geq 1$, one has the standard version of the dominated convergence theorem.

Corollary 2.3.12: (Lebesgue's dominated convergence theorem, or DCT). If $|f_n| \leq g$ a.e. (μ) for all $n \geq 1$, $\int g d\mu < \infty$ and $f_n \rightarrow f$ a.e. (μ) , then $f \in L^1(\Omega, \mathcal{F}, \mu)$, ① f integrable

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu \quad \text{and} \quad \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad (3.13)$$

Corollary 2.3.13: (The bounded convergence theorem, or BCT). Let $\mu(\Omega) < \infty$. If there exist a $0 < k < \infty$ such that $|f_n| \leq k$ a.e. (μ) and $f_n \rightarrow f$ a.e. (μ) , then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu \quad \text{and} \quad \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad (3.14)$$

Proof: Take $g(\omega) \equiv k$ for all $\omega \in \Omega$ in the previous corollary. $\square$

Proof of Theorem 2.3.11: By Fatou's lemma,

$$\int |f| d\mu \leq \liminf_{n \rightarrow \infty} \int |f_n| d\mu \leq \liminf_{n \rightarrow \infty} \int |g_n| d\mu = \int |g| d\mu < \infty.$$

$$\hookrightarrow \int \liminf (|f_n|) d\mu \downarrow \int |f| d\mu$$

$$|f_n| \leq g \Rightarrow g_n \geq 0 \text{ \& } g \geq 0$$

$$\int |f| d\mu < \infty$$

$$\leadsto \text{a.e. wrt } \mu$$

$$\leadsto \text{wrt } \mu(C) = 0$$

$$\int |f| d\mu < \infty$$

$$\text{as } n \rightarrow \infty$$

$$\hookrightarrow \int |f| d\mu < \infty \text{ \& } n < \infty$$

$$\therefore \text{ we can use m.c.f.a.}$$

$$\lim_{n \rightarrow \infty} \int |g_n| d\mu = \int |g| d\mu$$

$$\text{If } X_n \rightarrow X \text{ a.e. (P)} \\ \& |X_n| \leq Y \text{ a.e. (P)} \\ \& E|Y| < \infty, \text{ then} \\ \lim_{n \rightarrow \infty} E(X_n) = E(X) \uparrow \text{P.V.}$$

$$\lim \int f_n d\mu = \int \lim f_n d\mu \\ \text{If } |f_n| \leq g \in L_1$$

$$\textcircled{3} \Rightarrow \textcircled{2}$$

$$|f_n| \leq g \Rightarrow \int |f_n| d\mu \leq \int |g| d\mu$$

$$\int g d\mu = k \mu(\Omega) < \infty$$

$g \equiv k$
covers probabilities

when dealing w/ a bounded set of R.V., you can take the limit inside

note: $|f_n| \leq g_n \Rightarrow g_n \geq 0$

$$\Leftrightarrow -g_n \leq f_n \leq g_n \Rightarrow \begin{cases} f_n - g_n \geq 0 \\ f_n + g_n \geq 0 \end{cases} \Rightarrow h_n$$

Hence, f is integrable. For proving the second part, let $h_n = f_n + g_n$ and $\gamma_n = g_n - f_n, n \geq 1$. Then, $\{h_n\}_{n \geq 1}$ and $\{\gamma_n\}_{n \geq 1}$ are sequences of nonnegative integrable functions. By Fatou's lemma and (ii),

$$\begin{aligned} \int (f + g) d\mu &= \int \liminf_{n \rightarrow \infty} h_n d\mu \\ &\leq \liminf_{n \rightarrow \infty} \int h_n d\mu = \liminf_{n \rightarrow \infty} \int (g_n + f_n) d\mu \\ &= \liminf_{n \rightarrow \infty} \left[\int g_n d\mu + \int f_n d\mu \right] \\ &= \int g d\mu + \liminf_{n \rightarrow \infty} \int f_n d\mu. \end{aligned}$$

Handwritten notes: $\liminf (a_n + b_n) = a + \liminf (b_n)$ if $\lim a_n = a$; Fatou's ($h_n \geq 0$, nonneg); $f_n \rightarrow f$ & $g_n \rightarrow g$; $h_n = f_n + g_n$.

Similarly,

$$\int (g - f) d\mu \leq \int g d\mu - \limsup_{n \rightarrow \infty} \int f_n d\mu.$$

Handwritten notes: ≥ 0 ; $g_n \rightarrow g$ (given).

By Proposition 2.3.8, $\int (g \pm f) d\mu = \int g d\mu \pm \int f d\mu$. Hence,

$$\int f d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu$$

and

$$\limsup_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu$$

yielding that $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. For the last part, apply the above argument to f_n and g_n replaced by $\tilde{f}_n \equiv |f - f_n|$ and $\tilde{g}_n \equiv g_n + |f|$, respectively. $\square$

Handwritten notes: $\int f d\mu = \liminf_{n \rightarrow \infty} \int f_n d\mu = \limsup_{n \rightarrow \infty} \int f_n d\mu$; $f_n \rightarrow f$ $\tilde{f}_n = |f_n - f| \rightarrow 0$ $\Rightarrow \int \tilde{f}_n d\mu \rightarrow 0$.

end of DCT

Theorem 2.3.14: (An approximation theorem). Let μ_F be a Lebesgue-Stieltjes measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $f \in L^p(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$, $0 < p < \infty$. Then, for any $\delta > 0$, there exist a step function h and a continuous function g with compact support (i.e., g vanishes outside a bounded interval) such that

$$\int |f - h|^p d\mu < \delta, \quad (3.15)$$

$$\int |f - g|^p d\mu < \delta, \quad (3.16)$$

where a step function h is a function of the form $h = \sum_{i=1}^k c_i I_{A_i}$ with $k < \infty$, $c_1, c_2, \dots, c_k \in \mathbb{R}$ and $A_1, A_2, \dots, A_k$ being bounded disjoint intervals.

Proof: Let $f_n(\cdot) = f(\cdot) I_{B_n}(\cdot)$ where $B_n = \{x : |x| \leq n, |f(x)| \leq n\}$. By the DCT, for every $\epsilon > 0$, there exists an N_ϵ such that for all $n \geq N_\epsilon$,

$$\int |f - f_n|^p d\mu_F < \epsilon. \quad (3.17)$$

SKIP rest of 2.3

Since $|f_{N_\epsilon}(\cdot)| \leq N_\epsilon$ on $[-N_\epsilon, N_\epsilon]$, for any $\eta > 0$, there exists a simple function $\tilde{f}$ such that

$$\sup\{|f_{N_\epsilon}(x) - \tilde{f}(x)| : x \in \mathbb{R}\} < \eta. \quad (\text{cf. (3.7)}) \quad (3.18)$$

Next, using Problem 1.32 (b), one can show that for any $\eta > 0$, there exists a step function h (depending on η) such that

$$\int |\tilde{f}_{N_\epsilon} - h|^p d\mu_F < \eta. \quad (3.19)$$

Since $f - h = f - f_{N_\epsilon} + f_{N_\epsilon} - \tilde{f} + \tilde{f} - h$,

$$|f - h|^p \leq C_p \left(|f - f_{N_\epsilon}|^p + |f_{N_\epsilon} - \tilde{f}|^p + |\tilde{f} - h|^p \right),$$

where C_p is a constant depending only on p .

This in turn yields, from (3.17)–(3.19),

$$\int |f - h|^p d\mu_F \leq C_p (\epsilon + (\mu_F\{x : |x| \leq N_\epsilon\})\eta^p + \eta). \quad (3.20)$$

Given $\delta > 0$, choose $\epsilon > 0$ first and then $\eta > 0$ such that the right side of (3.20) above is less than δ .

Next, given any step function h and $\eta > 0$ there exists a continuous function g with compact support such that $\mu_F\{x : h(x) \neq g(x)\} < \eta$ (cf. Problem 1.32 (g)). Now (3.16) follows from (3.15). $\square$

Remark 2.3.9: The approximation (3.16) remains valid if g is restricted to the class of all infinitely differentiable functions with compact support. Further it remains valid for $0 < p < \infty$ for Lebesgue-Stieltjes measures on any Euclidean space.

Remark 2.3.10: The above approximation theorem fails for $p = \infty$. For example, consider the function $f(x) \equiv 1$ in $L^\infty(m)$.

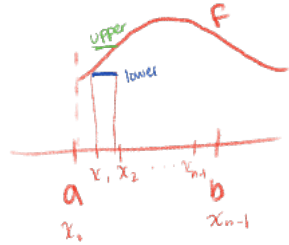
2.4 Riemann and Lebesgue integrals

VIDEO FOR THIS... FUCK

Let f be a real valued bounded function on a bounded interval $[a, b]$. Recall the definition of the *Riemann integral*. Let $P = \{x_0, x_1, \dots, x_n\}$ be a finite *partition* of $[a, b]$, i.e., $x_0 = a < x_1 < x_2 < \dots < x_{n-1} < x_n = b$ and $\Delta = \Delta(P) \equiv \max\{(x_{i+1} - x_i) : 0 \leq i \leq n-1\}$ be the *diameter* of P . Let $M_i = \sup\{f(x) : x_i \leq x \leq x_{i+1}\}$ and $m_i = \inf\{f(x) : x_i \leq x \leq x_{i+1}\}$, $i = 0, 1, \dots, n-1$.

Definition 2.4.1: The *upper-* and *lower-Riemann sums* of f w.r.t. the partition P are, respectively, defined as

$$U(f, P) \equiv \sum_{i=0}^{n-1} M_i \cdot (x_{i+1} - x_i) \quad (4.1)$$



and

$$L(f, P) \equiv \sum_{i=0}^{n-1} m_i \cdot (x_{i+1} - x_i). \quad (4.2)$$

It is easy to verify that if $Q = \{y_0, y_1, \dots, y_k\}$ is another partition satisfying $P \subset Q$, then $U(f, P) \geq U(f, Q) \geq L(f, Q) \geq L(f, P)$. Let $\mathcal{P}$ denote the collection of all finite partitions of $[a, b]$.

Definition 2.4.2: The *upper-Riemann integral* $\overline{\int} f$ is defined as

$$\overline{\int} f = \inf_{P \in \mathcal{P}} U(f, P) \quad (4.3)$$

and the *lower-Riemann integral* $\underline{\int} f$, by

$$\underline{\int} f = \sup_{P \in \mathcal{P}} L(f, P). \quad (4.4)$$

It can be shown (cf. Problem 2.23) that if $\{P_n\}_{n \geq 1}$ is any sequence of partitions such that $\Delta(P_n) \rightarrow 0$ as $n \rightarrow \infty$ and $P_n \subset P_{n+1}$ for each $n \geq 1$, then $U(P_n, f) \downarrow \overline{\int} f$ and $L(P_n, f) \uparrow \underline{\int} f$.

Definition 2.4.3: f is said to be *Riemann integrable* if

$$\overline{\int} f = \underline{\int} f. \quad (4.5)$$

The common value is denoted by $\oint_{[a,b]} f$.

Fix a sequence $\{P_n\}_{n \geq 1}$ of partitions such that $P_n \subset P_{n+1}$ for all $n \geq 1$ and $\Delta(P_n) \rightarrow 0$ as $n \rightarrow \infty$. Let $P_n = \{x_0 = a < x_{n1} < x_{n2} \dots < x_{nk_n} = b\}$. For $i = 0, 1, \dots, k_n - 1$, let

$$\begin{aligned} \phi_n(x) &\equiv \sup\{f(t) : x_i \leq t \leq x_{i+1}\}, \quad x \in [x_i, x_{i+1}] \\ \psi_n(x) &\equiv \inf\{f(t) : x_i \leq t \leq x_{i+1}\}, \quad x \in [x_i, x_{i+1}] \end{aligned}$$

and let $\phi_n(b) = \psi_n(b) = 0$. Then, ϕ_n and ψ_n are step functions on $[a, b]$ and hence, are Borel measurable. Further, since f is bounded, so are ϕ_n and ψ_n and hence are integrable on $[a, b]$ w.r.t. the Lebesgue measure m . The Lebesgue integrals of ϕ_n and ψ_n are given by $\int_{[a,b]} \phi_n dm = U(P_n, f)$ and $\int_{[a,b]} \psi_n dm = L(P_n, f)$.

It can be shown (Problem 2.24) that for all $x \notin \bigcup_{n \geq 1} P_n$, as $n \rightarrow \infty$,

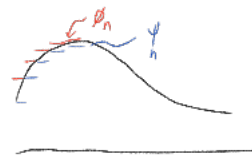
$$\phi_n(x) \rightarrow \phi(x) \equiv \lim_{\delta \downarrow 0} \sup\{f(y) : |y - x| < \delta\} \quad (4.6)$$

and

$$\psi_n(x) \rightarrow \psi(x) \equiv \lim_{\delta \downarrow 0} \inf\{f(y) : |y - x| < \delta\} \quad (4.7)$$

now $\int$ is for Lebesgue integral

$\int_a^b f(x) dx = \oint_{[a,b]} f$
usual notation Riemann integral local notation for Riemann integral (this section)



Thus, ϕ and ψ , being limits of Borel measurable functions (except possibly on a countable set), are Borel measurable. By the BCT (Corollary 2.3.13),

$$\overline{\int} f = \lim_{n \rightarrow \infty} \int \phi_n dm = \int \phi dm \quad \leftarrow \text{Lebesgue}$$

upper Riemann integral

and

$$\underline{\int} f = \lim_{n \rightarrow \infty} \int \psi_n dm = \int \psi dm.$$

lower Riemann integral

Thus, f is Riemann integrable on $[a, b]$, iff $\int \phi dm = \int \psi dm$, iff $\int (\phi - \psi) dm = 0$. Since $\phi(x) \geq f(x) \geq \psi(x)$ for all x , this holds iff $\phi = f = \psi$ a.e. $[m]$. It can be shown that f is continuous at x_0 iff $\phi(x_0) = f(x_0) = \psi(x_0)$ (Problem 2.8). Summarizing the above discussion, one gets the following theorem.

Theorem 2.4.1: Let f be a bounded function on a bounded interval $[a, b]$. Then f is Riemann integrable on $[a, b]$ iff f is continuous a.e. (m) on $[a, b]$. In this case, f is Lebesgue integrable on $[a, b]$ and the Lebesgue integral $\int_{[a,b]} f dm$ equals the Riemann integral $\oint_{[a,b]} f$, i.e., the two integrals coincide.

It should be noted that Lebesgue integrability need not imply Riemann integrability. For example, consider $f(x) \equiv I_{\mathbb{Q}_1}(x)$ where $\mathbb{Q}_1$ is the set of rationals in $[0, 1]$ (Problem 2.25).

The functions ϕ and ψ defined in (4.6) and (4.7) above are called, respectively, the *upper* and the *lower envelopes* of the function f . They are *semicontinuous* in the sense that for each $\alpha \in \mathbb{R}$, the sets $\{x : \phi(x) < \alpha\}$ and $\{x : \psi(x) > \alpha\}$ are open (cf. Problem 2.8).

Remark 2.4.1: The key difference in the definitions of Riemann and Lebesgue integrals is that in the former the *domain* of f is partitioned while in the latter the *range* of f is partitioned.

eg not clean
it's messy but
clean def
the Riemann
integral

Riemann
integrable $\Rightarrow$ Leb.
integrable

$$\oint_{[a,b]} f = \int_{[a,b]} f dm \quad \leftarrow \text{Leb. integral}$$

most cases

$$\int f dm = \int f dm = 0 \cdot m(\mathbb{Q}) + 1 \cdot m(\mathbb{Q}^c) = 0$$

$\mathbb{Q} \cap [0,1] = \mathbb{Q}$

For other cases, if $\oint f dm < \infty$, then $\oint f dm = \int f dm$

$\downarrow$
 $\oint_{[a,b]} f = \int_{[a,b]} f$

2.5 More on convergence (modes on convergence)

$$f_n \rightarrow f \quad \int f_n dm = \int f dm$$

Let $\{f_n\}_{n \geq 1}$ and f be measurable functions from a measure space $(\Omega, \mathcal{F}, \mu)$ to $\mathbb{R}$, the set of extended real numbers. There are several notions of convergence of $\{f_n\}_{n \geq 1}$ to f . The following two have been discussed earlier.

Definition 2.5.1: $\{f_n\}_{n \geq 1}$ converges to f pointwise if (everywhere)

$$\lim_{n \rightarrow \infty} f_n(\omega) = f(\omega) \quad \text{for all } \omega \text{ in } \Omega.$$

Definition 2.5.2: $\{f_n\}_{n \geq 1}$ converges to f almost everywhere (μ), denoted by $f_n \rightarrow f$ a.e. (μ), if there exists a set B in $\mathcal{F}$ such that $\mu(B) = 0$ and

$$\lim_{n \rightarrow \infty} f_n(\omega) = f(\omega) \quad \text{for all } \omega \in B^c. \quad (5.1)$$

Recall: $\{a_n\} \subset \mathbb{R}, a \in \mathbb{R}$
 $a_n \rightarrow a$ as $n \rightarrow \infty$
 $\Leftrightarrow$
 $\forall \varepsilon > 0 \exists N \in \mathbb{N} \text{ s.t.}$
 $|a_n - a| < \varepsilon \quad \forall n \geq N$

Important Ones

$$a_n = \mu(\{\omega: |f_n(\omega) - f(\omega)| > \epsilon\}) \rightarrow 0 = a$$

$$\mu(\{\omega: f(\omega) \neq f(\omega)\}) = 0$$

a.s(P) if $\mu = P$

Now consider some more notions of convergence.

Definition 2.5.3: $\{f_n\}_{n \geq 1}$ converges to f in measure (w.r.t. μ), denoted by $f_n \xrightarrow{m} f$, if for each $\epsilon > 0$,

not necessarily
Lebesgue measure

$$\lim_{n \rightarrow \infty} \mu(\{|f_n - f| > \epsilon\}) = 0. \quad \mu(\{|f_n - f| > \epsilon\}) \rightarrow 0 \quad (5.2)$$

when $\mu = P$ convergence
in probability

Definition 2.5.4: Let $0 < p < \infty$. Then, $\{f_n\}_{n \geq 1}$ converges to f in $L^p(\mu)$, denoted by $f_n \xrightarrow{L^p} f$, if $\int |f_n|^p d\mu < \infty$ for all $n \geq 1$, $\int |f|^p d\mu < \infty$ and

(converges in
pth moment)

$$\lim_{n \rightarrow \infty} \int |f_n - f|^p d\mu = 0. \quad f_n \in L^p \quad \forall n \geq 1 \quad f \in L^p \quad (5.3)$$

$\|f_n - f\|_p = \left(\int |f_n - f|^p d\mu \right)^{1/p}$ if $n \geq 1$
if $0 \leq n < 1$

Clearly, (5.3) is equivalent to $\|f_n - f\|_p \rightarrow 0$ as $n \rightarrow \infty$, where for any $\mathcal{F}$ -measurable function g and any $0 < p < \infty$,

$\mu = P$,
convergence in pth mean
 $E|X_n - X|^p \rightarrow 0$

$$\|g\|_p = \left(\int |g|^p d\mu \right)^{\min\{\frac{1}{p}, 1\}}. \quad (5.4)$$

For $p = 1$, this is also called *convergence in absolute deviation* and for $p = 2$, *convergence in mean square*.

Definition 2.5.5: $\{f_n\}_{n \geq 1}$ converges to f uniformly (over Ω) if

$$\lim_{n \rightarrow \infty} \sup\{|f_n(\omega) - f(\omega)| : \omega \in \Omega\} = 0. \quad (5.5)$$

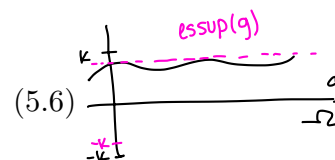
Definition 2.5.6: $\{f_n\}_{n \geq 1}$ converges to f in $L^\infty(\mu)$ if

$$\lim_{n \rightarrow \infty} \|f_n - f\|_\infty = 0, \quad (5.6)$$

where for any $\mathcal{F}$ -measurable function g on $(\Omega, \mathcal{F}, \mu)$,

$$\|g\|_\infty = \inf \{K : K \in (0, \infty), \mu(\{|g| > K\}) = 0\} = \text{ess sup}(g) \quad (5.7)$$

(essential supremum)



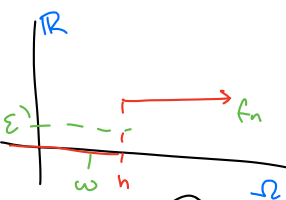
Definition 2.5.7: $\{f_n\}_{n \geq 1}$ converges to f nearly uniformly (μ) if for every $\epsilon > 0$, there exists a set $A \in \mathcal{F}$ such that $\mu(A) < \epsilon$ and on A^c , $f_n \rightarrow f$ uniformly, i.e., $\sup\{|f_n(\omega) - f(\omega)| : \omega \in A^c\} \rightarrow 0$ as $n \rightarrow \infty$.

The notion of convergence in Definition 2.5.7 is also called *almost uniform convergence* in some books (cf. Royden (1988)). The sequence $f_n \equiv nI_{[0, 1/n]}$ on $(\Omega = [0, 1], \mathcal{B}([0, 1]), m)$ converges to $f \equiv 0$ nearly uniformly but not in $L^\infty(m)$.

When μ is a probability measure, there is another useful notion of convergence, known as *convergence in distribution*, that is defined in terms of the induced measures $\{\mu f_n^{-1}\}_{n \geq 1}$ and μf^{-1} . This notion of convergence will be treated in detail in Chapter 9.

converges in
 L^p
 $\Rightarrow$ converges in L^q
where $p \geq q$

SKIP
less
used



check: infinite tail indicators
 $\Omega = \mathbb{R}, \mathcal{F} = \mathcal{B}(\mathbb{R})$
 $\mu = m(\text{Leb})$
 easy to check
 $f_n \rightarrow f (=0) \text{ a.e. } (\mu)$
 but, for $\varepsilon = 1/2 < 1$
 $\mu(|f_n - f| > \varepsilon) = \mu([0, h]) = h$
 $\rightarrow 0 \text{ as } n \rightarrow \infty$

Next, the connections between some of these notions of convergence are explored.

Theorem 2.5.1: Suppose that $\mu(\Omega) < \infty$. Then, $f_n \rightarrow f$ a.e. (μ) implies $f_n \rightarrow^m f$.

The proof is left as an exercise (Problem 2.26). The hypothesis that ' $\mu(\Omega) < \infty$ ' in Theorem 2.5.1 cannot be dispensed with as seen by taking ' $\mu(\Omega) < \infty$ ' in Theorem 2.5.1 cannot be dispensed with as seen by taking $f_n = I_{[n, \infty)}$ on $\mathbb{R}$ with Lebesgue measure. Also, $f_n \rightarrow^m f$ does not imply $f_n \rightarrow f$ a.e. (μ) (Problem 2.46), but the following holds.

Theorem 2.5.2: Let $f_n \rightarrow^m f$. Then, there exists a subsequence $\{n_k\}_{k \geq 1}$ such that $f_{n_k} \rightarrow f$ a.e. (μ) .

Choose $\varepsilon = \frac{1}{2^k}$, & choose $n_k \uparrow$.

Proof: Since $f_n \rightarrow^m f$, for each integer $k \geq 1$, there exists an n_k such that for all $n \geq n_k$

$$\mu\left(\underbrace{\{|f_n - f| > 2^{-k}\}}_{A_k}\right) < 2^{-k}. \quad \text{GIVEN} \quad (5.8)$$

W.l.o.g., that assume $n_{k+1} > n_k$ for all $k \geq 1$. Let $A_k = \{|f_{n_k} - f| > 2^{-k}\}$. By Corollary 2.3.5,

$$\int \left(\sum_{k=1}^{\infty} I_{A_k} \right) d\mu = \sum_{k=1}^{\infty} \int I_{A_k} d\mu = \sum_{k=1}^{\infty} \mu(A_k) = \sum_{k=1}^{\infty} \frac{1}{2^k} < \infty$$

which, by (5.8), is finite. Hence, by Proposition 2.3.10, $\sum_{k=1}^{\infty} I_{A_k} < \infty$ a.e. (μ) . Now observe that $\sum_{k=1}^{\infty} I_{A_k}(\omega) < \infty \Rightarrow |f_{n_k}(\omega) - f(\omega)| \leq 2^{-k}$ for all k large $\Rightarrow \lim_{k \rightarrow \infty} f_{n_k}(\omega) = f(\omega)$. Thus, $f_{n_k} \rightarrow f$ a.e. (μ) . $\square$

Remark 2.5.1: From the above result it follows that the extended dominated convergence theorem (Theorem 2.3.11) remains valid if convergence a.e. of $\{f_n\}_{n \geq 1}$ and of $\{g_n\}_{n \geq 1}$ are replaced by convergence in measure for both (Problem 2.37). can replace a.e. (μ) convergence w/ conv. in measure

Theorem 2.5.3: Let $\{f_n\}_{n \geq 1}, f$ be measurable functions on a measure space $(\Omega, \mathcal{F}, \mu)$. Let $f_n \rightarrow^{L^p} f$ for some $0 < p < \infty$. Then $f_n \rightarrow^m f$.

Proof: For each $\varepsilon > 0$, let $A_n = \{|f_n - f| \geq \varepsilon\}$, $n \geq 1$. Then $\lim_{n \rightarrow \infty} \mu(\{ \omega : |f_n(\omega) - f(\omega)| \geq \varepsilon \}) = 0$

$$\int |f_n - f|^p d\mu \geq \int_{A_n} |f_n - f|^p d\mu \geq \varepsilon^p \mu(A_n).$$

Since $f_n \rightarrow f$ in L^p , $\int |f_n - f|^p d\mu \rightarrow 0$ and hence, $\mu(A_n) \rightarrow 0$. $\square$

It turns out that $f_n \rightarrow^m f$ need not imply $f_n \rightarrow^{L^p} f$, even if $\{f_n : n \geq 1\} \cup \{f\}$ is contained in $L^p(\Omega, \mathcal{F}, \mu)$. For example, let $f_n = nI_{[0, \frac{1}{n}]}$ and $f \equiv 0$ on the Lebesgue space $([0, 1], \mathcal{B}([0, 1]), m)$, where m is the Lebesgue measure. Then $f_n \rightarrow^m f$ but $\int |f_n - f|^p \equiv 1$ for all $n \geq 1$. However, under

note that $|f_n - f| 1_{A_n} \geq \varepsilon 1_{A_n}$ a.e. (μ) [for e. $\varepsilon > 0$]

$$\Rightarrow |f_n - f|^p 1_{A_n} \geq \varepsilon^p 1_{A_n} \text{ a.e. } (\mu)$$

$$\Rightarrow \int |f_n - f|^p 1_{A_n} d\mu \geq \int \varepsilon^p 1_{A_n} d\mu \text{ a.e. } (\mu)$$

useful for proofs

$$f_n \rightarrow^m f$$

$$\mu(|f_n - f| > \varepsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\forall \varepsilon > 0 \quad \forall \varepsilon' > 0 \quad \exists N \geq 1 \text{ s.t.}$$

$$\mu(|f_n - f| > \varepsilon) < \varepsilon' \quad \forall n \geq N$$

$$\text{Choose } \varepsilon = \varepsilon' = \frac{1}{2^k} = 2^{-k}$$

$$N \text{ depends on } \varepsilon' \quad N_{\varepsilon'} \quad N_k$$

$$f_n \rightarrow^m f \Leftrightarrow \forall \varepsilon > 0$$

$$\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0$$

$$\forall \varepsilon > 0, \forall \varepsilon' > 0 \quad \exists N_{\varepsilon'} \geq 1 \text{ s.t. } \mu(|f_n - f| > \varepsilon) < \varepsilon' \quad \forall n \geq N_{\varepsilon'}$$

nonneg f_n has finite integral $\Rightarrow$ nonneg f_n is finite a.e.

for a lg. enough k , $I_{A_k} = 0$ since

$$\sum_{k=1}^{\infty} \int I_{A_k} d\mu < \infty \quad \& \quad I_{A_k} = 0 \text{ or } 1 \quad \forall k$$

mix L^p -convergence & measure convergence

converge in measure $\Rightarrow$ conv. in L^p
 counter example: "tall & thin"

In general, $f 1_{\{f \geq g\}} \geq g 1_{\{f \geq g\}}$ a.e. (μ) & $\therefore \int f 1_{\{f \geq g\}} d\mu \geq \int g 1_{\{f \geq g\}} d\mu$ a.e. (μ) (integrals are #, so a.e. (μ) doesn't make sense)

$\int_{\{f \geq g\}} f d\mu \geq \int_{\{f \geq g\}} g d\mu$ where $\int_{\{f \geq g\}}$

some additional conditions, convergence in measure does imply convergence in L^p . Here are two results in this direction.

note that useful in THIS class

Less restrictive, but only nonneg mble fns

Theorem 2.5.4: (Scheffe's theorem). Let $\{f_n\}_{n \geq 1}, f$ be a collection of nonnegative measurable functions on a measure space $(\Omega, \mathcal{F}, \mu)$. Let $f_n \rightarrow f$ a.e. (μ), $\int f_n d\mu \rightarrow \int f d\mu$ and $\int f d\mu < \infty$. Then

not used in approx. normal w/ binomial (binomial CLT)

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad f_n \rightarrow f \text{ in } L_1$$

Proof: Let $g_n = f - f_n$, $n \geq 1$. Since $f_n \rightarrow f$ a.e. (μ), both g_n^+ and g_n^- go to zero a.e. (μ). Further, $0 \leq g_n^+ \leq f$ and by hypothesis $\int f d\mu < \infty$. Thus, by the DCT, it follows that

$$\int g_n^+ d\mu \rightarrow 0.$$

Next, note that by hypothesis, $\int g_n d\mu \rightarrow 0$. Thus, $\int g_n^- d\mu = \int g_n^+ d\mu - \int g_n d\mu \rightarrow 0$ and hence, $\int |g_n| d\mu = \int g_n^+ d\mu + \int g_n^- d\mu \rightarrow 0$. $\square$

Corollary 2.5.5: Let $\{f_n\}_{n \geq 1}, f$ be probability density functions on a measure space $(\Omega, \mathcal{F}, \mu)$. That is, for all $n \geq 1$, $\int f_n d\mu = \int f d\mu = 1$ and $f_n, f \geq 0$ a.e. (μ). If $f_n \rightarrow f$ a.e. (μ), then

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

Remark 2.5.2: The above theorem and the corollary remain valid if the convergence of f_n to f a.e. (μ) is replaced by $f \xrightarrow{m} f$.

Corollary 2.5.6: Let $\{p_{nk}\}_{k \geq 1}$, $n = 1, 2, \dots$ and $\{p_k\}_{k \geq 1}$ be sequences of nonnegative numbers satisfying $\sum_{k=1}^{\infty} p_{nk} = 1 = \sum_{k=1}^{\infty} p_k$. Let $p_{nk} \rightarrow p_k$ as $n \rightarrow \infty$ for each $k \geq 1$. Then $\sum_{k=1}^{\infty} |p_{nk} - p_k| \rightarrow 0$.

Proof: Apply Corollary 2.5.5 with $\mu =$ the counting measure on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$. $\square$

A more general result in this direction that does not require f_n, f to be nonnegative involves the concept of *uniform integrability*. Let $\{f_\lambda : \lambda \in \Lambda\}$ be a collection of functions in $L^1(\Omega, \mathcal{F}, \mu)$. Then for each $\lambda \in \Lambda$, by the DCT and the integrability of f_λ ,

$$a_\lambda(t) \equiv \int_{\{|f_\lambda| > t\}} |f_\lambda| d\mu \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (5.9)$$

The notion of uniform integrability requires that the integrals $a_\lambda(t)$ go to zero uniformly in $\lambda \in \Lambda$ as $t \rightarrow \infty$.

$\{f_n : n \in \mathbb{N}\}$ or $\{X_n : n \in \mathbb{N}\}$ v.v. is UI

- each f_n/X_n is in L_1 ($f_n \in L_1$ or $E_n \in L_1$, i.e. $E|X_n| < \infty$)
- $\lim_{t \rightarrow \infty} \sup_{n \geq 1} \int |f_n| 1_{\{|f_n| > t\}} d\mu = 0$
- $\parallel \int |f_n| d\mu$
- $\int |f_n| 1_{\{|f_n| > t\}} d\mu$

woah,
 λ is a p cool letter

$$\text{or } \lim_{t \rightarrow \infty} \sup_{n \in \mathbb{N}} E \left[|x_n| \mathbb{1}_{\{|x_n| > t\}} \right] = 0$$

Definition 2.5.8: The collection of functions $\{f_\lambda : \lambda \in \Lambda\}$ in $L^1(\Omega, \mathcal{F}, \mu)$ is *uniformly integrable* (or *UI*, in short) if

$$\sup_{\lambda \in \Lambda} a_\lambda(t) \rightarrow 0 \text{ as } t \rightarrow \infty. \quad (5.10)$$

$a_\lambda(t) = \int_{|f_\lambda| > t} |f_\lambda| d\mu$

The following proposition summarizes some of the main properties of UI families of functions.

Proposition 2.5.7: Let $\{f_\lambda : \lambda \in \Lambda\}$ be a collection of μ -integrable functions on $(\Omega, \mathcal{F}, \mu)$.

(i) If Λ is finite, then $\{f_\lambda : \lambda \in \Lambda\}$ is UI. → e.g. collection has 2 fns: $\{f_1, f_2\}$

(ii) If $K \equiv \sup \left\{ \int |f_\lambda|^{1+\epsilon} d\mu : \lambda \in \Lambda \right\} < \infty$ for some $\epsilon > 0$, then $\{f_\lambda : \lambda \in \Lambda\}$ is UI. ↳ just $\sup \int |f_\lambda| d\mu : \lambda \in \Lambda < \infty$ is NOT sufficient for UI!

(iii) If $|f_\lambda| \leq g$ a.e. (μ) and $\int g d\mu < \infty$, then $\{f_\lambda : \lambda \in \Lambda\}$ is UI. → (iii') $|f_\lambda| \leq g_\lambda$ & $\{g_\lambda\}$ U.I. $\Rightarrow \{f_\lambda\}$ U.I.

(iv) If $\{f_\lambda : \lambda \in \Lambda\}$ and $\{g_\gamma : \gamma \in \Gamma\}$ are UI, then so is $\{f_\lambda + g_\gamma : \lambda \in \Lambda, \gamma \in \Gamma\}$. → (iv') $\sum \alpha_\lambda f_\lambda + \beta g_\gamma : \lambda \in \Lambda, \gamma \in \Gamma$ U.I. any lin comb.

(v) If $\{f_\lambda : \lambda \in \Lambda\}$ is UI and $\mu(\Omega) < \infty$, then

$$\sup_{\lambda \in \Lambda} \int |f_\lambda| d\mu < \infty. \quad (5.11)$$

Proof: By hypothesis, $a_\lambda(t) \equiv \int_{\{|f_\lambda| > t\}} |f_\lambda| d\mu \rightarrow 0$ as $t \rightarrow \infty$ for each λ . If Λ is finite this implies that $\sup \{a_\lambda(t) : \lambda \in \Lambda\} \rightarrow 0$ as $t \rightarrow \infty$. This proves (i).

To prove (ii), note that since $1 < |f_\lambda|/t$ on the set $\{|f_\lambda| > t\}$,

$$\sup_{\lambda \in \Lambda} a_\lambda(t) \leq \sup_{\lambda \in \Lambda} \int_{|f_\lambda| > t} |f_\lambda| \left[|f_\lambda|/t \right]^\epsilon d\mu \leq K t^{-\epsilon} \rightarrow 0 \text{ as } t \rightarrow \infty.$$

$a_\lambda(t) = \int_{|f_\lambda| > t} |f_\lambda| d\mu \leq \int_{|f_\lambda| > t} |f_\lambda| \left(\frac{|f_\lambda|}{t} \right)^\epsilon d\mu = \frac{1}{t^\epsilon} \int_{|f_\lambda| > t} |f_\lambda|^{1+\epsilon} d\mu \leq \frac{1}{t^\epsilon} K = \frac{K}{t^\epsilon}$

For (iii), note that for each $t \in \mathbb{R}$, the function $h_t(x) \equiv x I_{(t, \infty)}(x)$, $x \in \mathbb{R}$ is nondecreasing. Hence, by the integrability of g ,

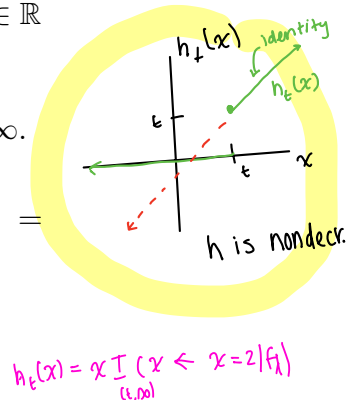
$$\sup_{\lambda \in \Lambda} a_\lambda(t) = \sup_{\lambda \in \Lambda} \int h_t(|f_\lambda|) d\mu \leq \int h_t(g) d\mu = \int_{\{g > t\}} g d\mu \rightarrow 0 \text{ as } t \rightarrow \infty.$$

(iii') $|f_\lambda| \leq g$ (f_λ bounded by g a.e. (μ))

To prove (iv), for $t > 0$, let $a(t) = \sup_{\lambda \in \Lambda} \int h_t(|f_\lambda|) d\mu$ and $b(t) = \sup_{\gamma \in \Gamma} \int h_t(|g_\gamma|) d\mu$. Then, for any $\lambda \in \Lambda$, and $\gamma \in \Gamma$,

$$\begin{aligned} \int h_t(|f_\lambda + g_\gamma|) d\mu &= \int_{\{|f_\lambda + g_\gamma| > t\}} |f_\lambda + g_\gamma| d\mu = \int h_t(|f_\lambda + g_\gamma|) d\mu \\ &\leq \int h_t(2 \max\{|f_\lambda|, |g_\gamma|\}) d\mu \end{aligned}$$

WTS: $\int h_t(|f_\lambda + g_\gamma|) d\mu \rightarrow 0$ as $t \rightarrow \infty$



$$h_t(x) = x \mathbb{I}_{(t, \infty)}(x) \quad (x=2|f_\lambda|)$$

$$\begin{aligned} &= \int h_t(2|f_\lambda|) I(|f_\lambda| \geq |g_\gamma|) d\mu + \int h_t(2|g_\gamma|) I(|f_\lambda| < |g_\gamma|) d\mu \leq \int h_t(2|f_\lambda|) d\mu + \int h_t(2|g_\gamma|) d\mu \\ &\leq 2 \int_{\{|f_\lambda| > t/2\}} |f_\lambda| d\mu + 2 \int_{\{|g_\gamma| > t/2\}} |g_\gamma| d\mu \\ &\leq 2[a(t/2) + b(t/2)]. \end{aligned}$$

By hypothesis, both $a(t)$ and $b(t) \rightarrow 0$ as $t \rightarrow \infty$, thus proving (iv).

Next consider (v). Since $\{f_\lambda\} : \lambda \in \Lambda$ is UI, there exists a $T > 0$ such that

$$\sup_{\lambda \in \Lambda} \int h_T(|f_\lambda|) d\mu \leq 1. \quad \begin{array}{l} \text{arbitrarily} \\ \text{chosen} \end{array} \quad \sup_{\lambda} (a_\lambda(T)) \rightarrow 0 \text{ as } T \rightarrow \infty$$

Hence,

$$\begin{aligned} \sup_{\lambda \in \Lambda} \int |f_\lambda| d\mu &= \sup_{\lambda \in \Lambda} \left\{ \int_{\{|f_\lambda| \leq T\}} |f_\lambda| d\mu + \int h_T(|f_\lambda|) d\mu \right\} \\ &\leq T\mu(\Omega) + 1 < \infty. \end{aligned}$$

NOTE: w/ UI fn, supremum of 1st moment is finite

Read up to ↓ here

This completes the proof of the proposition. $\square$

Remark 2.5.3: In the above proposition, part (ii) can be improved as follows: Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be nondecreasing and $\frac{\phi(x)}{x} \uparrow \infty$ as $x \uparrow \infty$. If $\sup_{\lambda \in \Lambda} \int \phi(|f_\lambda|) d\mu < \infty$, then $\{f_\lambda : \lambda \in \Lambda\}$ is UI (Problem 2.27). A converse to this result is true. That is, if $\{f_\lambda : \lambda \in \Lambda\}$ is UI then there exists such a function ϕ . Some examples of such ϕ 's are $\phi(x) = x^k$, $k > 1$, $\phi(x) = x(\log x)^\beta I(x > 1)$, $\beta > 0$, and $\phi(x) = \exp(\beta x)$, $\beta > 0$.

In part (v) of Proposition 2.5.7, (5.11) does not imply UI. For example, consider the sequence of functions $f_n = nI_{[0, \frac{1}{n}]}$, $n = 1, 2, \dots$ on $[0, 1]$. On the other hand, (5.11) with an additional condition becomes necessary and sufficient for UI.

Proposition 2.5.8: Let $f \in L^1(\Omega, \mathcal{F}, \mu)$. Then for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(A) < \delta \Rightarrow \int_A |f| d\mu < \epsilon$.

Proof: Fix $\epsilon > 0$. By the DCT, there exists a $t > 0$ such that $\int_{\{|f| > t\}} |f| d\mu < \epsilon/2$. Hence, for any $A \in \mathcal{F}$ with $\mu(A) \leq \delta \equiv \frac{\epsilon}{2t}$,

$$\begin{aligned} \int_A |f| d\mu &\leq \int_{A \cap \{|f| \leq t\}} |f| d\mu + \int_{\{|f| > t\}} |f| d\mu \\ &\leq t\mu(A) + \int_{\{|f| > t\}} |f| d\mu \\ &\leq \epsilon, \end{aligned}$$

proving the claim. $\square$

$\square$

UI + 2.5.10

The above proposition shows that for every $f \in L^1(\Omega, \mathcal{F}, \mu)$, the measure (cf. Corollary 2.3.6)

$$\nu_{|f|}(A) \equiv \int_A |f| d\mu \quad (5.12)$$

on $(\Omega, \mathcal{F})$ satisfies the condition that $\nu_{|f|}(A)$ is small if $\mu(A)$ is small, i.e., for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(A) < \delta \Rightarrow \nu_{|f|}(A) < \epsilon$. This property is referred to as the *absolute continuity* of the measure ν_f w.r.t. μ .

Definition 2.5.9: Given a family $\{f_\lambda : \lambda \in \Lambda\} \subset L^1(\Omega, \mathcal{F}, \mu)$, the measures $\{\nu_{|f_\lambda|} : \lambda \in \Lambda\}$ as defined in (5.12) above are *uniformly absolutely continuous* w.r.t. μ (or *u.a.c. (μ)*, in short) if for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\mu(A) < \delta \Rightarrow \sup \left\{ \nu_{|f_\lambda|}(A) : \lambda \in \Lambda \right\} < \epsilon.$$

Theorem 2.5.9: Let $\{f_\lambda : \lambda \in \Lambda\} \subset L^1(\Omega, \mathcal{F}, \mu)$ and $\mu(\Omega) < \infty$. Then, $\{f_\lambda : \lambda \in \Lambda\}$ is UI iff $\sup_{\lambda \in \Lambda} \int |f_\lambda| d\mu < \infty$ and $\{\nu_{|f_\lambda|}(\cdot) : \lambda \in \Lambda\}$ is u.a.c. (μ).

Proof: Let $\{f_\lambda : \lambda \in \Lambda\}$ be UI. Then, since $\mu(\Omega) < \infty$, L_1 boundedness of $\{f_\lambda : \lambda \in \Lambda\}$ follows from Proposition 2.5.7 (v). To establish u.a.c. (μ), fix $\epsilon > 0$. By UI, there exists an N such that

$$\sup_{\lambda \in \Lambda} \int_{\{|f_\lambda| > N\}} |f_\lambda| d\mu < \epsilon/2.$$

Let $\delta = \frac{\epsilon}{2N}$ and let $A \in \mathcal{F}$ be such that $\mu(A) < \delta$. Then, as in the proof of Proposition 2.5.8 above,

$$\sup_{\lambda \in \Lambda} \int_A |f_\lambda| d\mu \leq N\mu(A) + \epsilon/2 < \epsilon.$$

Conversely, suppose $\{f_\lambda : \lambda \in \Lambda\}$ is L^1 bounded and u.a.c. (μ). Then, for every $\epsilon > 0$, there exists a $\delta_\epsilon > 0$ such that

$$\sup_{\lambda \in \Lambda} \int_A |f_\lambda| d\mu < \epsilon \text{ if } \mu(A) < \delta_\epsilon. \quad (5.13)$$

Also, for any nonnegative f in $L_1(\Omega, \mathcal{F}, \mu)$ and $t > 0$, $\int f d\mu \geq \int_{\{f > t\}} f d\mu \geq t\mu(\{f \geq t\})$, which implies that

$$\mu(\{f \geq t\}) \leq \frac{\int f d\mu}{t}.$$

(This is known as *Markov's inequality* – see Chapter 3). Hence, it follows that

$$\sup_{\lambda \in \Lambda} \mu(\{|f_\lambda| \geq t\}) \leq \left[\sup_{\lambda \in \Lambda} \int |f_\lambda| d\mu \right] / t. \quad (5.14)$$

Now given $\epsilon > 0$, choose T_ϵ such that $\left[\sup_{\lambda \in \Lambda} \int |f_\lambda| d\mu \right] / T_\epsilon < \delta_\epsilon$ where δ_ϵ is as in (5.13). Then, by (5.14), it follows that

$$\sup_{\lambda \in \Lambda} \int_{\{|f_\lambda| \geq T_\epsilon\}} |f_\lambda| d\mu < \epsilon,$$

i.e., $\{f_\lambda : \lambda \in \Lambda\}$ is UI. $\square$

Theorem 2.5.10: Let $(\Omega, \mathcal{F}, \mu)$ be a measure space with $\mu(\Omega) < \infty$, and let $\{f_n : n \geq 1\} \subset L^1(\Omega, \mathcal{F}, \mu)$ be such that $f_n \rightarrow f$ a.e. (μ) and f is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. If $\{f_n : n \geq 1\}$ is UI, then f is integrable and

DCT holds

only VI theorem

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

If $\mu(\Omega) < \infty$ & $f_n \rightarrow f$ a.e. (μ) & $\{f_n\}$ is UI then $f \in L_1$ & $f_n \rightarrow f$ in $L_1(\mu)$

$\Rightarrow$ Thm 2.5.10 holds

Remark 2.5.4: In view of Proposition 2.5.7 (iii), Theorem 2.5.10 yields convergence of $\int f_n d\mu$ to $\int f d\mu$ under weaker conditions than the DCT, provided $\mu(\Omega) < \infty$. However, even under the restriction $\mu(\Omega) < \infty$, UI of $\{f_n\}_{n \geq 1}$ is a sufficient, but *not* a necessary condition for convergence of $\int f_n d\mu$ to $\int f d\mu$ (Problem 2.28). In the special case where f_n 's are non-negative (and $\mu(\Omega) < \infty$), $\int f_n d\mu \rightarrow \int f d\mu < \infty$ if and only if $\{f_n\}_{n \geq 1}$ are UI (Problem 2.29). On the other hand, when $\mu(\Omega) = +\infty$, UI is no longer sufficient to guarantee the convergence of $\int f_n d\mu$ to $\int f d\mu$ (Problem 2.30). Thus, the notion of UI is useful mainly for *finite* measures and, in particular, probability measures.

Check: $f_n \rightarrow f$ in measure can replace $f_n \xrightarrow{a.e.} f$ in Thm 2.5.10

Proof: By Proposition 2.5.7 and Fatou's lemma,

$$\int (\liminf_{n \rightarrow \infty} |f_n|) d\mu = \int |f| d\mu \leq \liminf_{n \rightarrow \infty} \int |f_n| d\mu \leq \sup_{n \geq 1} \int |f_n| d\mu < \infty$$

and hence f is integrable.

Next for $n \in \mathbb{N}, t \in (0, \infty)$, define the functions $g_n = |f_n - f|$, $\underline{g}_{n,t} = g_n I(|g_n| \leq t)$ and $\bar{g}_{n,t} = g_n I(|g_n| > t)$. Since $\{f_n\}_{n \geq 1}$ is UI and f is integrable, by Proposition 2.5.7 (iv), $\{g_n\}_{n \geq 1}$ is UI. Hence, for any $\epsilon > 0$, there exists $t_\epsilon > 0$ such that

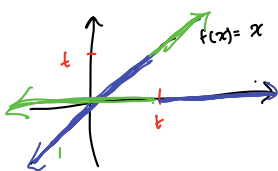
$$\sup_{n \geq 1} \int \bar{g}_{n,t} d\mu < \epsilon \quad \text{for all } t \geq t_\epsilon. \quad (5.15)$$

Next note that for any $t > 0$, since $f_n \rightarrow f$ a.e. (μ) , $\underline{g}_{n,t} \rightarrow 0$ a.e. (μ) , and $|\underline{g}_{n,t}| \leq t$ and $\int(t) d\mu = t\mu(\Omega) < \infty$. Hence, by the DCT,

$$\lim_{n \rightarrow \infty} \int \underline{g}_{n,t} d\mu = 0 \quad \text{for all } t > 0. \quad (5.16)$$

$$\bar{h}_t(x) = x I_{(-t, \infty)}(x)$$

$$\underline{h}_t(x) = x I_{(-\infty, t]}(x)$$



check: $g_n \rightarrow 0$ a.e.
 h is cont @ 0
then $h(g_n) \rightarrow h(0)$ a.e.

READ

Let $f_K(x) = f(x)I_{[a,b]}(x)I_{\{|f| \leq K\}}(x)$. Then $f_K : \mathbb{R} \rightarrow \mathbb{R}$ is bounded, $\langle \mathcal{M}_{\mu_F^*}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable and zero for $|x| > K$. Consider now the following claim: *For every $\epsilon > 0$, there exists a continuous function $g : [a, b] \rightarrow \mathbb{R}$ such that*

$$\mu(\{x : f_K(x) \neq g(x), a \leq x \leq b\}) < \frac{\epsilon}{2}. \quad (5.17)$$

Clearly this implies that

$$\mu(\{x : a \leq x \leq b, f(x) \neq g(x)\}) < \epsilon. \quad (5.18)$$

Fix $\delta > 0$. Now for each $n \in \mathbb{Z}$, take $[a, b] = [n, n+1]$, $\epsilon = \frac{\delta}{2^{|n|+2}}$, apply (5.18) and call the resulting continuous function g_n . Let $\tilde{g}$ be a continuous function from $\mathbb{R} \rightarrow \mathbb{R}$ such that $\mu(\{x : n \leq x \leq n+1, \tilde{g}(x) \neq g_n(x)\}) < \frac{\delta}{2^{|n|+2}}$. This can be done by setting $\tilde{g}(x) = g_n(x)$ for $n \leq x \leq (n+1) - \delta_n$ and linear on $[(n+1) - \delta_n, n+1]$ for some $0 < \delta_n < \frac{\delta}{2^{|n|+3}}$. Then

$$\begin{aligned} \mu(\{x : f(x) \neq \tilde{g}(x)\}) &\leq \sum_{n=-\infty}^{\infty} \mu(\{x : n \leq x \leq n+1, f(x) \neq \tilde{g}(x)\}) \\ &\leq \sum_{n=-\infty}^{\infty} \mu(\{x : n \leq x \leq n+1, f(x) \neq g_n(x)\}) \\ &\quad + \sum_{n=-\infty}^{\infty} \mu(\{x : n \leq x \leq n+1, g_n(x) \neq \tilde{g}(x)\}) \\ &< 2 \sum_{n=-\infty}^{\infty} \frac{\delta}{2^{|n|+2}} \leq 2\delta. \end{aligned}$$

So it suffices to establish (5.17). Since $f_K : \mathbb{R} \rightarrow \mathbb{R}$ is bounded and $\langle \mathcal{M}_{\mu_F^*}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, for each $\epsilon > 0$, there exists a simple function $s(x) \equiv \sum_{i=1}^k c_i I_{A_i}(x)$, with $A_i \subset [a, b]$, $A_i \in \mathcal{M}_{\mu_F^*}$, $\{A_i : 1 \leq i \leq k\}$ are disjoint, $\mu(A_i) < \infty$, and $c_i \in \mathbb{R}$ for $i = 1, \dots, k$, such that $|f(x) - s(x)| < \frac{\epsilon}{4}$ for all $a \leq x \leq b$. By Theorem 1.3.4, for each A_i and $\eta > 0$, there exist a finite number of open disjoint intervals $I_{ij} = (a_{ij}, b_{ij})$, $j = 1, \dots, n_i$ such that

$$\mu\left(A_i \triangle \bigcup_{j=1}^{n_i} I_{ij}\right) < \frac{\eta}{2k}.$$

Now as in Problem 1.32 (g), there exists a continuous function g_{ij} such that

$$\mu(g_{ij}^{-1}\{1\} \triangle I_{ij}) < \frac{\eta}{kn_i}, \quad j = 1, 2, \dots, n_i, \quad i = 1, 2, \dots, k.$$

Let

$$g_i \equiv \sum_{j=1}^{n_i} g_{ij}, \quad 1 \leq i \leq k.$$

Then $\mu(A_i \triangle g_i^{-1}\{1\}) < \frac{\eta}{k}$. Let $g = \sum_{i=1}^k c_i g_i$. Then $\mu(\{s \neq g\}) < \eta$. Hence for every $\epsilon > 0$, $\eta > 0$, there is a continuous function $g_{\epsilon, \eta} : [a, b] \rightarrow \mathbb{R}$ such that

$$\mu(\{x : a \leq x \leq b, |f_K(x) - g_{\epsilon, \eta}(x)| > \epsilon\}) < \eta.$$

Now for each $n \geq 1$, let

$$h_n(\cdot) \equiv g_{\frac{1}{2^n}, \frac{1}{2^n}}(\cdot) \quad \text{and} \quad A_n = \left\{x : a \leq x \leq b, |f_K(x) - h_n(x)| > \frac{1}{2^n}\right\}.$$

Then, $\mu(A_n) < \frac{1}{2^n}$ and hence

$$\sum_{n=1}^{\infty} \mu(A_n) < \infty.$$

By the MCT, this implies that $\int_{[a, b]} (\sum_{n=1}^{\infty} I_{A_n}) d\mu < \infty$ and hence

$$\sum_{n=1}^{\infty} I_{A_n} < \infty \quad \text{a.e.} \quad \mu.$$

Thus $h_n \rightarrow f_K$ a.e. μ on $[a, b]$. By Egorov's theorem for any $\epsilon > 0$, there is a set $A_\epsilon \in \mathcal{B}([a, b])$ such that

$$\mu(A_\epsilon^c) < \epsilon/2 \quad \text{and} \quad h_n \rightarrow f_K \quad \text{uniformly on } A_\epsilon.$$

By the inner regularity (Corollary 1.3.5) of μ , there is a compact set $D \subset A_\epsilon$ such that

$$\mu(A_\epsilon \setminus D) < \epsilon/2.$$

Since $h_n \rightarrow f_K$ uniformly on A_ϵ , f_K is continuous on A_ϵ and hence on D . It can be shown that there exists a continuous function $g : [a, b] \rightarrow \mathbb{R}$ such that $g = f_K$ on D (Problem 2.8 (e)). A more general result extending a continuous function defined on a closed set to the whole space is known as Tietze's extension theorem (see Munkres (1975)). Thus $\mu(\{x : a \leq x \leq b, f_K(x) \neq g(x)\}) < \epsilon$. This completes the proof of (5.17) and hence that of the proposition. $\square$

Remark 2.5.5: (*Littlewood's principles*). As pointed out in Section 1.3, Theorems 1.3.4, 2.5.11, and 2.5.12 constitute J. E. Littlewood's three principles: every $\mathcal{M}_{\mu_F^*}$ measurable set is nearly a finite union of intervals; every a.e. convergent sequence is nearly uniformly convergent; and every $\mathcal{M}_{\mu_F^*}$ -measurable function is nearly continuous.

2.6 Problems

2.1 Let Ω_i , $i = 1, 2$ be two nonempty sets and $T : \Omega_1 \rightarrow \Omega_2$ be a map. Then for any collection $\{A_\alpha : \alpha \in I\}$ of subsets of Ω_2 , show that

$$T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) = \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$$

$$\text{and } T^{-1}\left(\bigcap_{\alpha \in I} A_\alpha\right) = \bigcap_{\alpha \in I} T^{-1}(A_\alpha).$$

Further, $(T^{-1}(A))^c = T^{-1}(A^c)$ for all $A \subset \Omega_2$. (These are known as *de Morgan's laws*.)

2.2 Let $\{A_i\}_{i \geq 1}$ be a collection of disjoint sets in a measurable space $(\Omega, \mathcal{F})$.

- (a) Let $\{g_i\}_{i \geq 1}$ be a collection of $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable functions from Ω to $\mathbb{R}$. Show that $\sum_{i=1}^{\infty} I_{A_i} g_i$ converges on $\mathbb{R}$ and is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
- (b) Let $\mathcal{G} \equiv \sigma\{A_i : i \geq 1\}$. Show that $h : \Omega \rightarrow \mathbb{R}$ is $\langle \mathcal{G}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable iff $g(\cdot)$ is constant on each A_i .

2.3 Let $f, g : \Omega \rightarrow \mathbb{R}$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Set

$$h(\omega) = \frac{f(\omega)}{g(\omega)} I(g(\omega) \neq 0), \quad \omega \in \Omega.$$

Verify that h is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

2.4 Let $g : \Omega \rightarrow \bar{\mathbb{R}}$ be such that for every $r \in \mathbb{R}$, $g^{-1}((-\infty, r]) \in \mathcal{F}$. Show that g is $\langle \mathcal{F}, \mathcal{B}(\bar{\mathbb{R}}) \rangle$ -measurable.

2.5 Prove Proposition 2.1.6.

(**Hint:** Show that

$$\sigma\{f_\lambda : \lambda \in \Lambda\} = \bigcup_{L \in \mathcal{C}} \sigma\{f_\lambda : \lambda \in L\}$$

where $\mathcal{C}$ is the collection of all countable subsets of Λ .)

2.6 Let $X_i, i = 1, 2, 3$ be random variables on a probability space $(\Omega, \mathcal{F}, P)$. Consider the random equation (in $t \in \mathbb{R}$):

$$X_1(\omega)t^2 + X_2(\omega)t + X_3(\omega) = 0. \quad (6.1)$$

- (a) Show that $A \equiv \{\omega \in \Omega : \text{Equation (6.1) has two distinct real roots}\} \in \mathcal{F}$.
- (b) Let $T_1(\omega)$ and $T_2(\omega)$ denote the two roots of (6.1) on A . Let

$$f_i(w) = \begin{cases} T_i(w) & \text{on } A \\ 0 & \text{on } A^c \end{cases},$$

$i = 1, 2$. Show that (f_1, f_2) is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ -measurable.

2.7 Let $M \equiv ((X_{ij})), 1 \leq i, j \leq k$, be a (random) matrix of random variables X_{ij} defined on a probability space $(\Omega, \mathcal{F}, P)$.

- (a) Show that $Y_1 \equiv \det(M)$ (the determinant of M) and $Y_2 \equiv \text{tr}(M)$ (the trace of M) are both $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.
- (b) Show also that $Y_3 \equiv$ the largest eigenvalue of $M'M$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, where M' is the transpose of M .

(**Hint:** Use the result that $Y_3 = \sup_{x \neq 0} \frac{(x' M' M x)}{(x' x)}.$)

2.8 Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $\bar{f}(x) = \inf_{\delta > 0} \sup_{|y-x| < \delta} f(y)$ and $\underline{f}(x) = \sup_{\delta > 0} \inf_{|y-x| < \delta} f(y)$, $x \in \mathbb{R}$.

- (a) Show that for any $t \in \mathbb{R}$,

$$\{x : \bar{f}(x) < t\}$$

is open and hence, $\bar{f}$ is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

- (b) Show that for any $t > 0$,

$$\{x : \bar{f}(x) - \underline{f}(x) < t\} \equiv \bigcup_{r \in \mathbb{Q}} \{x : \bar{f}(x) < t + r, \underline{f}(x) > r\}$$

and hence is open.

- (c) Show that f is continuous at some x_0 in $\mathbb{R}$ iff $\bar{f}(x_0) = \underline{f}(x_0)$.
- (d) Show that the set $C_f \equiv \{x : f(\cdot) \text{ is continuous at } x\}$ is a G_δ set, i.e., an intersection of a countable number of open sets, and hence, C_f is a Borel set.
- (e) Let D be a closed set in $\mathbb{R}$. Let $f : D \rightarrow \mathbb{R}$ be continuous on D . Show that there exists a $g : \mathbb{R} \rightarrow \mathbb{R}$ continuous such that $g = f$ on D .

(**Hint:** Note that D^c is open in $\mathbb{R}$ and hence it can be expressed as a countable union of disjoint open intervals $\{I_j = (a_j, b_j) : 1 \leq j \leq k \leq \infty\}$. Note that $a_j, b_j \in D$ for all j except for possibly the j 's for which $a_j = -\infty$ or $b_j = +\infty$. Let

$$g(x) \equiv \begin{cases} f(x) & \text{if } x \in D \\ f(a_j) + \frac{(x-a_j)}{(b_j-a_j)}(f(b_j) - f(a_j)) & \text{if } x \in (a_j, b_j), \\ & a_j, b_j \in D \\ f(b_j) & \text{if } a_j = -\infty, \\ & x \in (a_j, b_j) \\ f(a_j) & \text{if } b_j = \infty, \\ & x \in (a_j, b_j). \end{cases}$$

Now verify that g has the required properties.)

2.9 Prove Proposition 2.2.1 using Problem 2.1.

2.10 (a) Show that for any $x \in \mathbb{R}$ and any random variable X with cdf $F_X(\cdot)$, $P(X < x) = F_X(x-) \equiv \lim_{y \uparrow x} F_X(y)$.

(b) Show that $F_c(\cdot)$ in (2.5) is continuous.

2.11 Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be nondecreasing.

(a) Show that for $x \in \mathbb{R}$, $F(x-) \equiv \lim_{y \uparrow x} F(y)$ and $F(x+) = \lim_{y \downarrow x} F(y)$ exist and satisfy $F(x-) \leq F(x) \leq F(x+)$.

(b) Let $D \equiv \{x : F(x+) - F(x-) > 0\}$. Show that D is at most countable.

(**Hint:** Show that

$$D = \bigcup_{n \geq 1} \bigcup_{r \geq 1} D_{n,r},$$

where $D_{n,r} = \{x : |x| \leq n, F(x+) - F(x-) > \frac{1}{r}\}$ and that each $D_{n,r}$ is finite.)

2.12 Suppose that (i) and (iii) of Proposition 2.2.3 hold. Show that for any a_1 in $\mathbb{R}$ and $-\infty < a_2 \leq b_2 < \infty$, $F(a_2, a_1) \leq F(b_2, a_1)$ and $F(a_1, a_2) \leq F(a_1, b_2)$, (i.e., F is monotone coordinatewise).

2.13 Let $F : \mathbb{R}^k \rightarrow \mathbb{R}$ be such that:

(a) for $x_1 = (x_{11}, x_{12}, \dots, x_{1k})$ and $x_2 = (x_{21}, x_{22}, \dots, x_{2k})$ with $x_{1i} \leq x_{2i}$ for $i = 1, 2, \dots, k$,

$$\Delta F(x_1, x_2) \equiv \sum_{a \in A} (-1)^{s(a)} F(a) \geq 0,$$

where $A \equiv \{a = (a_1, a_2, \dots, a_k) : a_i \in \{x_{1i}, x_{2i}\}, i = 1, 2, \dots, k\}$ and for a in A , $s(a) = |\{i : a_i = x_{1i}, i = 1, 2, \dots, k\}|$ is the number of indices i for which $a_i = x_{1i}$.

(b) For each $i = 1, 2, \dots, k$, $\lim_{x_{i1} \downarrow -\infty} F(x_i) = 0$.

Let $\mathcal{C}_k$ be the semialgebra of sets of the form $\{A : A = A_1 \times \dots \times A_k, A_i \in \mathcal{C} \text{ for all } 1 \leq i \leq k\}$ where $\mathcal{C}$ is the semialgebra in $\mathbb{R}$ defined in (3.7). Set $\mu_F(A) \equiv \Delta F(x_1, x_2)$ if $A = (x_{11}, x_{21}] \times (x_{12}, x_{22}] \times \dots \times (x_{1k}, x_{2k}]$ is bounded and set $\mu_F(A) = \lim_{n \rightarrow \infty} \mu(A \cap J_n)$, where $J_n = (-n, n]^k$.

(i) Show that F is coordinatewise monotone, i.e., if $x = (x_1, \dots, x_k)$, $y = (y_1, \dots, y_k)$ and $x_i \leq y_i$ for every $i = 1, \dots, k$, then

$$F(y) \geq F(x).$$

- (ii) Show that $\mathcal{C}$ is a semialgebra and μ_F is a measure on $\mathcal{C}$ by using the Heine-Borel theorem as in Problem 1.22 and 1.23.

2.14 Let $m(\cdot)$ denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be the map $T(x) = x^2$. Evaluate the induced measure $mT^{-1}(A)$ of the set A , where

- (a) $A = [0, t], t > 0$.
- (b) $A = (-\infty, 0)$.
- (c) $A = \{1, 2, 3, \dots\}$.
- (d) $A = \bigcup_{i=1}^{\infty} (i^2, (i + \frac{1}{i^2})^2)$.
- (e) $A = \bigcup_{i=1}^{\infty} (i^2, (i + \frac{1}{i})^2)$.

2.15 Consider the probability space $((0, 1), \mathcal{B}((0, 1)), m)$, where $m(\cdot)$ is the Lebesgue measure.

- (a) Let Y_1 be the random variable $Y_1(x) \equiv \sin 2\pi x$ for $x \in (0, 1)$. Find the cdf of Y_1 .
- (b) Let Y_2 be the random variable $Y_2(x) \equiv \log x$ for $x \in (0, 1)$. Find the cdf of Y_2 .
- (c) Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a cdf. For $0 < x < 1$, let

$$\begin{aligned} F_1^{-1}(x) &= \inf\{y : y \in \mathbb{R}, F(y) \geq x\} \\ F_2^{-1}(x) &= \sup\{y : y \in \mathbb{R}, F(y) \leq x\}. \end{aligned}$$

Let Z_i be the random variable defined by

$$Z_i = F_i^{-1}(x) \quad 0 < x < 1, \quad i = 1, 2.$$

- (i) Find the cdf of $Z_i, i = 1, 2$.

(Hint: Verify using the right continuity of F that for any $0 < x < 1, t \in \mathbb{R}, F(t) \geq x \Leftrightarrow F_1^{-1}(x) \leq t$.)

- (ii) Show also that $F_1^{-1}(\cdot)$ is left continuous and $F_2^{-1}(\cdot)$ is right continuous.

- 2.16 (a) Let $(\Omega, \mathcal{F}_1, \mu)$ be a σ -finite measure space. Let $T : \Omega \rightarrow \mathbb{R}$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Show, by an example, that the induced measure μT^{-1} need not be σ -finite.
- (b) Let $(\Omega_i, \mathcal{F}_i)$ be measurable spaces for $i = 1, 2$ and let $T : \Omega_1 \rightarrow \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable. Show that any measure μ on $(\Omega_1, \mathcal{F}_1)$ is σ -finite if μT^{-1} is σ -finite on $(\Omega_2, \mathcal{F}_2)$.

2.17 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f : \Omega \rightarrow [0, \infty]$ be such that it admits two representations

$$f = \sum_{i=1}^k c_i I_{A_i} \quad \text{and} \quad f = \sum_{j=1}^{\ell} d_j I_{B_j},$$

where $c_i, d_j \in [0, \infty]$, and A_i and $B_j \in \mathcal{F}$ for all i, j . Show that

$$\sum_{i=1}^k c_i \mu(A_i) = \sum_{j=1}^{\ell} d_j \mu(B_j).$$

(**Hint:** Express A_i and B_j as finite unions of a common collection of disjoint sets in $\mathcal{F}$.)

2.18 (a) Prove Proposition 2.3.1.

(b) In the proof of Proposition 2.3.2, verify that $D_n \uparrow D$.

(c) Verify Remark 2.3.1.

(**Hint:** Let

$$\begin{aligned} A_n &= \{\omega : f_{n-1}(\omega) \geq f_n(\omega), g_{n+1}(\omega) \geq g_n(\omega)\} \\ A &= \left(\bigcap_{n \geq 1} A_n \right) \cap \left\{ \omega : \lim_{n \rightarrow \infty} g_n(\omega) = g(\omega), \right. \\ &\quad \left. \lim_{n \rightarrow \infty} f_n(\omega) = f(\omega) \right\} \end{aligned}$$

and $\tilde{f}_n = f_n I_A$, $\tilde{g}_n = g_n I_A$. Verify that $\mu(A^c) = 0$ and apply Proposition 2.3.2 to $\{\tilde{f}_n\}_{n \geq 1}$ and $\{\tilde{g}_n\}_{n \geq 1}$.)

2.19 (a) Verify that $f_n(\cdot)$ defined in (3.7) satisfies $f_n(\omega) \uparrow f(\omega)$ for all ω in Ω .

(b) Verify that the sequence $\{h_n\}_{n \geq 1}$ of Remark 2.3.3 satisfies $\lim_{n \rightarrow \infty} h_n = f$ a.e. (μ), and $\lim_{n \rightarrow \infty} \int h_n d\mu = M$.

2.20 Apply Corollary 2.3.5 to show that for any collection $\{a_{ij} : i, j \in \mathbb{N}\}$ of nonnegative numbers,

$$\sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right) = \sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right).$$

2.21 Let $g : \mathbb{R} \rightarrow \mathbb{R}$.

(a) Recall that $\lim_{t \rightarrow \infty} g(t) = L$ for some L in $\mathbb{R}$ if for every $\epsilon > 0$, there exists a $T_\epsilon < \infty$ such that $t \geq T_\epsilon \Rightarrow |g(t) - L| < \epsilon$. Show that $\lim_{t \rightarrow \infty} g(t) = L$ for some L in $\mathbb{R}$ iff $\lim_{n \rightarrow \infty} g(t_n) = L$ for every sequence $\{t_n\}_{n \geq 1}$ with $\lim_{n \rightarrow \infty} t_n = \infty$.

- (b) Formulate and prove a similar result when $\lim_{t \rightarrow a} g(t) = L$ for some $a, L \in \mathbb{R}$.

2.22 Let $\{f_t : t \in \mathbb{R}\} \subset L^1(\Omega, \mathcal{F}, \mu)$.

- (a) (*The continuous version of the MCT*). Suppose that $f_t \uparrow f$ as $t \uparrow \infty$ a.e. (μ) and for each t , $f_t \geq 0$ a.e. (μ) . Show that

$$\int f_t d\mu \uparrow \int f d\mu.$$

- (b) (*The continuous version of the DCT*). Suppose there exists a nonnegative $g \in L^1(\Omega, \mathcal{F}, \mu)$ such that for each t , $|f_t| \leq g$ a.e. (μ) and as $t \rightarrow \infty$, $f_t \rightarrow f$ a.e. (μ) . Then $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\int |f_t - f| d\mu \rightarrow 0$ and hence, $\int f_t d\mu \rightarrow \int f d\mu$, as $t \rightarrow \infty$.

2.23 Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded where $-\infty < a < b < \infty$. Let $\{P_n\}_{n \geq 1}$ be a sequence of partitions such that $\Delta(P_n) \rightarrow 0$. Show that as $n \rightarrow \infty$,

$$U(P_n, f) \rightarrow \overline{\int} f \quad \text{and} \quad L(P_n, f) \rightarrow \underline{\int} f,$$

where $\overline{\int} f$ and $\underline{\int} f$ are as defined in (4.3) and (4.4), respectively.

(**Hint:** Given $\epsilon > 0$, fix a partition $P = \{x_0 = a < x_1 < \dots < x_k = b\}$ such that $\overline{\int} f < U(P, f) < \overline{f} + \epsilon$. Let $\delta = \min_{0 \leq i \leq k-1} (x_{i+1} - x_i)$. Choose n large such that the diameter $\Delta(P_n) < \delta$. Verify that

$$U(P_n, f) < U(P, f) + kB\Delta(P_n)$$

where $B = \sup\{|f(x)| : a \leq x \leq b\}$ and conclude that $\overline{\lim}_n U(P_n, f) \leq \overline{\int} f + \epsilon$.)

2.24 Establish (4.6) and (4.7).

(**Hint:** Show that for every x and any $\epsilon > 0$, $\phi_n(x) \leq \phi(x) + \epsilon$ for all n large and that for $x \notin \bigcup_{n \geq 1} P_n$, $\phi_n(x) \geq \phi(x)$ for all n .)

2.25 If $f(x) = I_{\mathbb{Q}_1}(x)$ where $\mathbb{Q}_1 = \mathbb{Q} \cap [0, 1]$, $\mathbb{Q}$ being the set of all rationals, then show that for any partition P , $U(P, f) = 1$ and $L(P, f) = 0$.

2.26 Establish Theorem 2.5.1.

(**Hint:** Verify that

$$\begin{aligned} D &\equiv \{\omega : f_j(\omega) \not\rightarrow f(\omega)\} \\ &= \bigcup_{r \geq 1} \bigcap_{n \geq 1} \bigcup_{j \geq n} A_{jr}, \end{aligned}$$

where $A_{jr} = \{|f_j - f| > \frac{1}{r}\}$. Show that since $\mu(D) = 0$ and $\mu(\Omega) < \infty$, $\mu(D_{rn}) \rightarrow 0$ as $n \rightarrow \infty$ for each $r \in \mathbb{N}$, where $D_{rn} = \bigcup_{j \geq n} A_{jr}$.)

- 2.27 Let $\phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be nondecreasing and $\frac{\phi(x)}{x} \uparrow \infty$ as $x \uparrow \infty$. Also, let $\{f_\lambda : \lambda \in \Lambda\}$ be a subset of $L^1(\Omega, \mathcal{F}, \mu)$. Show that if $\sup_{\lambda \in \Lambda} \int \phi(|f_\lambda|) d\mu < \infty$, then $\{f_\lambda : \lambda \in \Lambda\}$ is UI.
- 2.28 Let μ be the Lebesgue measure on $([-1, 1], \mathcal{B}([-1, 1]))$. For $n \geq 1$, define $f_n(x) = nI_{(0, n^{-1})}(x) - nI_{(-n^{-1}, 0)}(x)$ and $f(x) \equiv 0$ for $x \in [-1, 1]$. Show that $f_n \rightarrow f$ a.e. (μ) and $\int f_n d\mu \rightarrow \int f d\mu$ but $\{f_n\}_{n \geq 1}$ is not UI.
- 2.29 Let $\{f_n : n \geq 1\} \cup \{f\} \subset L^1(\Omega, \mathcal{F}, \mu)$.
- (a) Show that $\int |f_n - f| d\mu \rightarrow 0$ iff $f_n \xrightarrow{m} f$ and $\int |f_n| d\mu \rightarrow \int |f| d\mu$.
 - (b) Show further that if $\mu(\Omega) < \infty$ then the above two are equivalent to $f_n \xrightarrow{m} f$ and $\{f_n\}$ UI.
- 2.30 For $n \geq 1$, let $f_n(x) = n^{-1/2}I_{(0, n)}(x)$, $x \in \mathbb{R}$, and let $f(x) = 0$, $x \in \mathbb{R}$. Let m denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Show that $f_n \rightarrow f$ a.e. (m) and $\{f_n\}_{n \geq 1}$ is UI, but $\int f_n dm \not\rightarrow \int f dm$.
- 2.31 (*Computing integrals w.r.t. the Lebesgue measure*). Let $f \in L^1(\mathbb{R}, \mathcal{M}_m, m)$ where $(\mathbb{R}, \mathcal{M}_m, m)$ is the real line with Lebesgue σ -algebra, and Lebesgue measure, i.e., $m = \mu_F^*$ where $F(x) \equiv x$. The definition of $\int f dm$ as $\int f^+ dm - \int f^- dm$ involves computing $\int f^+ dm$ and $\int f^- dm$ which in turn is given in terms of approximating by integrals of simple nonnegative functions. This is not a very practical procedure. For f that happens to be continuous a.e. and bounded on finite intervals, one can compute the Riemann integral of f over finite intervals and pass to the limit. Justify the following steps:

- (a) Let f be continuous a.e. and bounded on finite intervals and $f \in L^1(\mathbb{R}, \mathcal{M}_m, m)$. Show that for $-\infty < a < b < \infty$, $f \in L^1([a, b], \mathcal{M}_m, m)$ and

$$\int_{[a, b]} f dm = \oint_{[a, b]} f(x) dx,$$

where the right side denotes the Riemann integral of f on $[a, b]$.

- (b) If, in addition, $f \in L^1(\mathbb{R}, \mathcal{M}_m, m)$, then

$$\int_{\mathbb{R}} f dm = \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow +\infty}} \int_{[a, b]} f dm.$$

- (c) If f is continuous a.e. and $f \in L^1(\mathbb{R}, \mathcal{M}_m, m)$, then

$$\int_{\mathbb{R}} f dm = \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow +\infty \\ c \rightarrow \infty}} \int_{[a, b]} \phi_c(f) dm$$

where $\phi_c(f) = f(x)I(|f(x)| \leq c) + cI(f(x) > c) - cI(f(x) < -c)$.

(d) Apply the above procedure to compute $\int_{\mathbb{R}} f dm$ for

(i) $f(x) = \frac{1}{1+x^2}$,

(ii) $f(x) = e^{-x^2/2}$,

(iii) $f(x) = e^{+x}e^{-x^2/2}$.

2.32 (a) Let $a \in \mathbb{R}$. Show that if a_1, a_2 are nonnegative such that $a = a_1 - a_2$ then $a_1 \geq a^+$, $a_2 \geq a^-$ and $a_1 - a^+ = a_2 - a^-$.

(b) Let $f = f_1 - f_2$ where f_1, f_2 are nonnegative and are in $L^1(\Omega, \mathcal{F}, \mu)$. Show that $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\int f d\mu = \int f_1 d\mu - \int f_2 d\mu$.

2.33 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Let $f : \Omega \times (a, b) \rightarrow \mathbb{R}$ be such that for each $a < t < b$, $f(\cdot, t) \in L^1(\Omega, \mathcal{F}, \mu)$.

(a) Suppose for each $a < t < b$,

(i) $\lim_{h \rightarrow 0} f(\omega, t+h) = f(\omega, t)$ a.e. (μ) .

(ii) $\sup_{|h| \leq 1} |f(\omega, t+h)| \leq g_1(\omega, t)$, where $g_1(\cdot, t) \in L^1(\Omega, \mathcal{F}, \mu)$.

Show that $\phi(t) \equiv \int_{\Omega} f(\omega, t) d\mu$ is continuous on (a, b) .

(b) Suppose for each $a < t < b$.

(i) $\lim_{h \rightarrow 0} \frac{f(\omega, t+h) - f(\omega, t)}{h} = g_2(\omega, t)$ exists a.e. (μ) ,

(ii) $\sup_{0 \leq |h| \leq 1} \left| \frac{f(\omega, t+h) - f(\omega, t)}{h} \right| \leq G(\omega, t)$ a.e. (μ) ,

(iii) $G(\omega, t) \in L^1(\Omega, \mathcal{F}, \mu)$.

Show that

$$\phi(t) \equiv \int_{\Omega} f(\omega, t) d\mu$$

is differentiable on (a, b) .

(**Hint:** Use the continuous version of DCT (cf. Problem 2.22).)

2.34 Let $A \equiv ((a_{ij}))$ be an infinite matrix of real numbers. Suppose that for each j , $\lim_{i \rightarrow \infty} a_{ij} = a_j$ exists in $\mathbb{R}$ and $\sup_i |a_{ij}| \leq b_j$, where $\sum_{j=1}^{\infty} b_j < \infty$.

(a) Show by an application of the DCT that

$$\lim_{i \rightarrow \infty} \sum_{j=1}^{\infty} |a_{ij} - a_j| = 0.$$

(b) Show the same directly, i.e., without using the DCT.

- 2.35 Using the above problem or otherwise, show that for any sequence $\{x_n\}_{n \geq 1}$ with $\lim_{n \rightarrow \infty} x_n = x$ in $\mathbb{R}$,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x_n}{n}\right)^n = \sum_{j=0}^{\infty} \frac{x^j}{j!} \equiv \exp(x).$$

- 2.36 (a) Let $\{f_n\}_{n \geq 1} \subset L^1(\Omega, \mathcal{F}, \mu)$ such that $f_n \rightarrow 0$ in $L^1(\mu)$. Show that $\{f_n\}_{n \geq 1}$ is UI.

- (b) Let $\{f_n\}_{n \geq 1} \subset L^p(\Omega, \mathcal{F}, \mu)$, $0 < p < \infty$, such that $\mu(\Omega) < \infty$, $\{|f_n|^p\}_{n \geq 1}$ is UI and $f_n \xrightarrow{m} f$. Show that $f \in L^p(\mu)$ and $f_n \rightarrow f$ in $L^p(\mu)$.

"in measure"
& a.e. can be
used interchangeably

- 2.37 (a) Show that for a sequence of real numbers $\{x_n\}_{n \geq 1}$, $\lim_{n \rightarrow \infty} x_n = x \in \mathbb{R}$ holds iff every subsequence $\{x_{n_j}\}_{j \geq 1}$ of $\{x_n\}_{n \geq 1}$ has a further subsequence $\{x_{n_{j_k}}\}_{k \geq 1}$ such that $\lim_{k \rightarrow \infty} x_{n_{j_k}} = x$.

- (b) Use (a) and Theorem 2.5.2 to show that the extended DCT (Theorem 2.3.11) is valid if the a.e. convergence of $\{f_n\}_{n \geq 1}$ and $\{g_n\}_{n \geq 1}$ is replaced by convergence in measure.

- 2.38 Let $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ be a Lebesgue-Stieltjes measure space generated by a $F : \mathbb{R} \rightarrow \mathbb{R}$ nondecreasing. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be $\mathcal{M}_{\mu_F^*}$ -measurable such that $|f| < \infty$ a.e. (μ_F^*) . Show that for every $k \in \mathbb{N}$ and for every $\epsilon, \eta \in (0, \infty)$, there exists a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) = 0$ for $|x| > k$ and $\mu_F^*(\{x : |x| \leq k, |f(x) - g(x)| > \eta\}) < \epsilon$.

(Hint: Complete the following:

STEP I: For all $\epsilon > 0$, there exists $M_{k, \epsilon} \in (0, \infty)$ such that $\mu_F^*(\{x : |x| \leq k, |f(x)| > M_{k, \epsilon}\}) < \epsilon$.

STEP II: For $\eta > 0$, there exists a simple function $s(\cdot)$ such that $\mu_F^*(\{x : |x| \leq k, |f(x)| \leq M_{k, \epsilon}, |f(x) - s(x)| > \eta\}) = 0$.

STEP III: For $\delta > 0$, there exists a continuous function $g(\cdot)$ such that $g \equiv 0$ for $|x| > k$ and $\mu_F^*(\{x : |x| \leq k, s(x) \neq g(x)\}) < \delta$.)

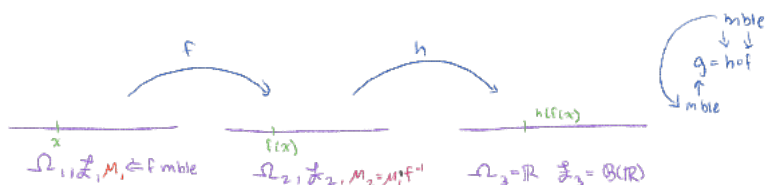
- 2.39 Recall from Corollary 2.3.6 that for $g \in L^1(\Omega, \mathcal{F}, \mu)$ and nonnegative $\nu_g(A) \equiv \int_A g d\mu$ is a measure. Next for any $\mathcal{F}$ -measurable function h , show that $h \in L^1(\nu_g)$ iff $h \cdot g \in L^1(\mu)$ and $\int h d\nu_g = \int h g d\mu$.

(Hint: Verify first for h simple and nonnegative, next for h nonnegative, and finally for any h .)

- 2.40 Prove the BCT, Corollary 2.3.13, using Egorov's theorem (Theorem 2.5.11).

- 2.41 Deduce the DCT from the BCT with the notation as in Corollary 2.3.12.

(Hint: Apply the BCT to the measure space $(\Omega, \mathcal{F}, \nu_g)$ and functions $h_n = \frac{f_n}{g} I(g > 0)$, $h = \frac{f}{g} I(g > 0)$ where ν_g is as in Problem 2.39.)



Proof:

• he started w/ h from Ω_2
• you could start w/ f from Ω_1

Proofs w/ integrals

no idea $\Rightarrow$ start w/ indicators

• simple fn
• lim simple fn = nonneg mble fn
• split mble fn into pos & neg pos-neg

$$\begin{aligned} \int_{\Omega_1} g d\mu_1 &= \int_{\Omega_2} h d\mu_2 \\ \text{att: } \int_{\Omega_1} h(f(\omega)) d\mu_1(\omega) &= \int_{\Omega_2} h(z) d(\mu_1 \circ f^{-1})(z) \end{aligned}$$

note:

identity fns
are always
mble/integrable

His favorite

2.42 (Change of variables formula). Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be two measurable spaces. Let $f : \Omega_1 \rightarrow \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -measurable, $h : \Omega_2 \rightarrow \mathbb{R}$ be $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$ -measurable, and μ_1 be a measure on $(\Omega_1, \mathcal{F}_1)$. Show that $g \equiv h \circ f$, i.e., $g(\omega) \equiv h(f(\omega))$ for $\omega \in \Omega_1$ is in $L^1(\mu)$ iff $h(\cdot) \in L^1(\Omega_2, \mathcal{F}_2, \mu_2)$ where $\mu_2 = \mu_1 f^{-1}$ iff $I(\cdot) \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3 \equiv \mu_2 h^{-1})$ where $I(\cdot)$ is the identity function in $\mathbb{R}$, i.e., $I(x) \equiv x$ for all $x \in \mathbb{R}$ and also that

$$\int_{\Omega_1} g d\mu_1 = \int_{\Omega_2} h d\mu_2 = \int_{\mathbb{R}} x d\mu_3.$$

2.43 Let $\phi(x) \equiv \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ be the standard $N(0, 1)$ pdf on $\mathbb{R}$. Let $\{(\mu_n, \sigma_n)\}_{n \geq 1}$ be a sequence in $\mathbb{R} \times \mathbb{R}_+$. Suppose $\mu_n \rightarrow \mu$ and $\sigma_n \rightarrow \sigma$ as $n \rightarrow \infty$. Let $f_n(x) = \frac{1}{\sigma_n} \phi\left(\frac{x - \mu_n}{\sigma_n}\right)$, $f(x) = \frac{1}{\sigma} \phi\left(\frac{x - \mu}{\sigma}\right)$ and $\nu_n(A) = \int_A f_n dm$, $\nu(A) = \int_A f dm$ for any Borel set A in $\mathbb{R}$, where $m(\cdot)$ is the Lebesgue measure on $\mathbb{R}$. Using Scheffe's theorem, verify that, as $n \rightarrow \infty$, $\nu_n(\cdot) \rightarrow \nu(\cdot)$ uniformly on $\mathcal{B}(\mathbb{R})$ and that for any $h : \mathbb{R} \rightarrow \mathbb{R}$, bounded and Borel measurable,

$$\int h d\nu_n \rightarrow \int h d\nu.$$

2.44 Let $f_n(x) = c_n \left(1 - \frac{x}{n}\right)^n I_{[0, n]}(x)$, $x \in \mathbb{R}$, $n \geq 1$.

- Find c_n such that $\int f_n dm = 1$.
- Show that $\lim_{n \rightarrow \infty} f_n(x) \equiv f(x)$ exists for all x in $\mathbb{R}$ and that f is a pdf
- For $A \in \mathcal{B}(\mathbb{R})$, let

$$\nu_n(A) \equiv \int_A f_n dm \quad \text{and} \quad \nu(A) = \int_A f dm.$$

Show that $\nu_n \rightarrow \nu$ uniformly on $\mathcal{B}(\mathbb{R})$.

2.45 Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f : \Omega \rightarrow \mathbb{R}$ be $\mathcal{F}$ -measurable. Suppose that $\int_{\Omega} |f| d\mu < \infty$ and D is a closed set in $\mathbb{R}$ such that for all $B \in \mathcal{F}$ with $\mu(B) > 0$,

$$\frac{1}{\mu(B)} \int_B f d\mu \in D.$$

Show that $f(\omega) \in D$ a.e. μ .

(**Hint:** Show that for $x \notin D$, there exists $r > 0$ such that $\mu\{\omega : |f(\omega) - x| < r\} = 0$.)

- 2.46 Find a sequence of nonnegative continuous functions $\{f_n\}_{n \geq 1}$ on $[0, 1]$ such that $\int_{[0, 1]} f_n dm \rightarrow 0$ but $\{f_n(x)\}_{n \geq 1}$ does not converge for any x .

(**Hint:** Let for $m \geq 1$, $1 \leq k \leq m$, $g_{m,k} = I_{\left[\frac{k-1}{m}, \frac{k}{m}\right]}$ and $\{f_n\}_{n \geq 1}$ be a reordering of $\{g_{n,k} : 1 \leq k \leq n, n \geq 1\}$.)

- 2.47 Let $\{f_n\}_{n \geq 1}$ be a sequence of continuous functions from $[0, 1]$ to $[0, 1]$ such that $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$ for all $0 \leq x \leq 1$. Show that $\int_0^1 f_n(x) dx \rightarrow 0$ as $n \rightarrow \infty$ (where the integral is the Riemann integral) by two methods: one by using BCT and one without using BCT. Show also that if μ is a finite measure on $([0, 1], \mathcal{B}([0, 1]))$, then $\int_{[0, 1]} f_n d\mu \rightarrow 0$ as $n \rightarrow \infty$.

- 2.48 (*Invariance of Lebesgue measure under translation and reflection.*) Let $m(\cdot)$ be Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

- (a) For any $E \in \mathcal{B}(\mathbb{R})$ and $c \in \mathbb{R}$, define $-E \equiv \{x : -x \in E\}$ and $E + c \equiv \{y : y = x + c, x \in E\}$. Show that

$$\begin{aligned} m(-E) &= m(E) \quad \text{and} \\ m(E + c) &= m(E) \end{aligned}$$

for all $E \in \mathcal{B}(\mathbb{R})$ and $c \in \mathbb{R}$.

- (b) For any $f \in L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ and $c \in \mathbb{R}$, let $\tilde{f}(x) \equiv f(-x)$ and $f_c(x) \equiv f(x + c)$. Show that

$$\int \tilde{f} dm = \int f_c dm = \int f dm.$$